4.20.11 - 200-Pin DDR2 SDRAM Unbuffered SO-DIMM Design Specification

PC2-6400/PC2-5300/PC2-4200/PC2-3200 DDR2 Unbuffered SO-DIMM

Reference Design Specification

Revision 2.5

July, 2008

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1. Product Description

This reference specification defines the electrical and mechanical requirements for the PC2-6400 memory module, a 200-pin, 400MHz clock (800 MT/s data rate), 64-bit wide, Unbuffered Synchronous Double Data Rate 2 (DDR2) DRAM Small Outline Dual In-Line Memory Module (DDR2 SDRAM SO-DIMMs). It also defines slower versions, the PC2-5300, PC2-4200 and PC2-3200 with a 333MHz 267MHz and 200MHz clock (667 MT/s, 533 MT/s and 400 MT/s data rate) DDR2 SDRAMs. These DDR2 SDRAM SO-DIMMs are intended for use as main memory when installed in systems such as mobile personal computers.

Reference design examples are included which provide an initial basis for Unbuffered SO-DIMM designs. Any modifications to these reference designs must meet all system timing, signal integrity and thermal requirements for 333MHz clock rate support. Other designs are acceptable, and all Unbuffered DDR2 SO-DIMM implementations must use simulations and lab verification to ensure proper timing requirements and signal integrity in the design.

Product Family Attributes

Attribute:	Values:	Notes:
SO-DIMM Organization	x 64	
Dimensions (nominal)	30.0 mm high, 67.60 mm wide / MO-224 variation CB 25.4 mm high, 67.60 mm wide / MO-224 variation AB	
SO-DIMM Types Supported	Unbuffered	
Pin Count	200	
SDRAMs Supported	256 Mb, 512 Mb, 1 Gb, 2 Gb, 4 Gb	
Capacity	128 MB, 256 MB, 512 MB, 1 GB, 2GB, 4 GB, 8 GB	
Serial Presence Detect	Consistent with JEDEC latest Rev.	
Voltage Options, Nominal	1.8 V V_{DD} 1.8 V $V_{DD}Q$ 1.8 V to 3.3 V $V_{DD}SPD$	1, 2
Interface	SSTL_18	

Note 1: $V_{DD}SPD$ is not tied to V_{DD} or $V_{DD}Q$ on the DDR2 SO-DIMM.

Note 2: SO-DIMMs that include an optional temperature sensor may require a restricted VDDSPD operating voltage range for proper operation of the temperature sensor. Refer to the thermal sensor specification for details regarding the supported voltage range. All other functions of the SO-DIMM SPD are supported across the full VDDSPD range.

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Raw Card Summary

Raw Card	Number of DDR2 SDRAMs	SDRAM Organization	Number of Ranks	Comments
A, G	8	x16	2	G is low profile version(maximum DRAM size, W x L) 11.0 mm x 13.0mm
B	8	x8	1	
C, H	4	x16	1	H is low profile version(maximum DRAM size, W x L) 11.0 mm x 13.0mm
D	16	x8	2	Uses stacked memory
E	16	x8	2	Planar (maximum DRAM size, W x L) 12.0 mm x 11.5 mm
F	16	x8	2	Planar (maximum DRAM size, W x L) 11.0 mm x 13.65 mm

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2. Environmental Requirements

PC2-6400 DDR2 SDRAM Unbuffered SO-DIMMs are intended for use in mobile computing environments that have limited capacity for heat dissipation.

Absolute Maximum Ratings

Symbol	Parameter	Rating	Units	Notes
T_{OPR}	Operating Temperature (ambient)	0 to +65	°C	1

1. Stresses greater than those listed may cause permanent damage to the device. This is a stress rating only, and device functional operation at or above the conditions indicated is not implied. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

3. Architecture

Pin Description

CK[1:0]	Clock Inputs, positive line	2	DQ[63:0]	Data Input/Output	64
$\overline{CK}$[1:0]	Clock inputs, negative line	2	DM[7:0]	Data Masks	8
CKE[1:0]	Clock Enables	2	DQS[7:0]	Data strobes	8
$\overline{RAS}$	Row Address Strobe	1	$\overline{DQS}$[7:0]	Data strobes complement	8
$\overline{CAS}$	Column Address Strobe	1			
$\overline{WE}$	Write Enable	1	TEST	Logic Analyzer specific test pin (No connect on SO-DIMM)	1
$\overline{S}$[1:0]	Chip Selects	2			
A[9:0],A[11:15]	Address Inputs	15	V_{DD}	Core and I/O Power	12
A10/AP	Address Input/Autoprecharge	1	V_{SS}	Ground	57
BA[2:0]	SDRAM Bank Address	3	V_{REF}	Input/Output Reference	1
ODT[1:0]	On-die termination control	2	V_{DD}SPD	SPD and TS Power	1
SCL	Serial Presence Detect (SPD) and Thermal sensor Clock Input	1	$\overline{Event}$ Pin	Reserved for optional hardware temperature sensing	1
SDA	SPD and TS Data Input/Output	1	NC	Reserved for future use	3
SA[1:0]	SPD and TS address	2		**Total:**	**200**

Note: 4 rank SO-DIMM specific signals($\overline{RESET}$, $\overline{S}$2, $\overline{S}$3) are not included in this table.

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Input/Output Functional Description

Symbol	Type	Polarity	Function
CK0/$\overline{CK0}$, CK1/$\overline{CK1}$	Input	Cross point	The system clock inputs. All address and command lines are sampled on the cross point of the rising edge of CK and falling edge of $\overline{CK}$. A Delay Locked Loop (DLL) circuit is driven from the clock inputs and output timing for read operations is synchronized to the input clock.
CKE[1:0]	Input	Active High	Activates the DDR2 SDRAM CK signal when high and deactivates the CK signal when low. By deactivating the clocks, CKE low initiates the Power Down mode or the Self Refresh mode.
$\overline{S}$[1:0]	Input	Active Low	Enables the associated DDR2 SDRAM command decoder when low and disables the command decoder when high. When the command decoder is disabled, new commands are ignored but previous operations continue. Rank 0 is selected by $\overline{S0}$; Rank 1 is selected by $\overline{S1}$.
$\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$	Input	Active Low	When sampled at the cross point of the rising edge of CK and falling edge of $\overline{CK}$ and $\overline{CAS}$, $\overline{RAS}$, and $\overline{WE}$ define the operation to be executed by the SDRAM.
BA[2:0]	Input	—	Selects which DDR2 SDRAM internal bank of four or eight is activated.
ODT[1:0]	Input	Active High	Asserts on-die termination for DQ, DM, DQS, and $\overline{DQS}$ signals if enabled via the DDR2 SDRAM mode register.
A[9:0], A10/AP, A[15:11]	Input	—	During a Bank Activate command cycle, defines the row address when sampled at the cross point of the rising edge of CK and falling edge of $\overline{CK}$. During a Read or Write command cycle, defines the column address when sampled at the cross point of the rising edge of CK and falling edge of $\overline{CK}$. In addition to the column address, AP is used to invoke autoprecharge operation at the end of the burst read or write cycle. If AP is high, autoprecharge is selected and BA0-BAn defines the bank to be precharged. If AP is low, autoprecharge is disabled. During a Precharge command cycle, AP is used in conjunction with BA0-BAn to control which bank(s) to precharge. If AP is high, all banks will be precharged regardless of the state of BA0-BAn inputs. If AP is low, then BA0-BAn are used to define which bank to precharge.
DQ[63:0]	In/Out	—	Data Input/Output pins.
DM[7:0]	Input	Active High	The data write masks, associated with one data byte. In Write mode, DM operates as a byte mask by allowing input data to be written if it is low but blocks the write operation if it is high. In Read mode, DM lines have no effect.
DQS[7:0], $\overline{DQS}$[7:0]	In/Out	Cross point	The data strobes, associated with one data byte, sourced with data transfers. In Write mode, the data strobe is sourced by the controller and is centered in the data window. In Read mode, the data strobe is sourced by the DDR2 SDRAMs and is sent at the leading edge of the data window. $\overline{DQS}$ signals are complements, and timing is relative to the crosspoint of respective DQS and $\overline{DQS}$. If the module is to be operated in single ended strobe mode, all $\overline{DQS}$ signals must be tied on the system board to VSS and DDR2 SDRAM mode registers programmed appropriately.
V_{DD}, V_{DD}SPD, V_{SS}	Supply	—	Power supplies for core, I/O, Serial Presence Detect, Thermal sensor, and ground for the module.
V_{REF}	Supply	—	Reference voltage for SSTL18 inputs.
SDA	In/Out	—	This is a bidirectional pin used to transfer data into or out of the SPD EEPROM or Thermal sensor. A resistor must be connected from the SDA bus line to V_{DD}SPD on the system planar to act as a pull up.
SCL	Input	—	This signal is used to clock data into and out of the SPD EEPROM and Thermal sensor.
SA[1:0]	Input	—	Address pins used to select the Serial Presence Detect base address.
TEST	In/Out	—	The TEST pin is reserved for bus analysis tools and is not connected on normal memory modules (SO-DIMMs).
$\overline{Event}$	Wire-OR Out	Active Low	The optional EVENT pin is reserved for use to flag critical module temperatures and is used in conjuction with a SPD temperture sensing option.

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DDR2 SDRAM SO-DIMM Pinout

Pin #	Front Side	Pin #	Back Side	Pin #	Front Side	Pin #	Back Side	Pin #	Front Side	Pin #	Back Side	Pin #	Front Side	Pin #	Back Side
1	V_{REF}	2	V_{SS}	51	DQS2	52	DM2	101	A1	102	A0	151	DQ42	152	DQ46
3	V_{SS}	4	DQ4	53	V_{SS}	54	V_{SS}	103	V_{DD}	104	V_{DD}	153	DQ43	154	DQ47
5	DQ0	6	DQ5	55	DQ18	56	DQ22	105	A10/AP	106	BA1	155	V_{SS}	156	V_{SS}
7	DQ1	8	V_{SS}	57	DQ19	58	DQ23	107	BA0	108	$\overline{RAS}$	157	DQ48	158	DQ52
9	V_{SS}	10	DM0	59	V_{SS}	60	V_{SS}	109	$\overline{WE}$	110	$\overline{S0}$	159	DQ49	160	DQ53
11	$\overline{DQS0}$	12	V_{SS}	61	DQ24	62	DQ28	111	V_{DD}	112	V_{DD}	161	V_{SS}	162	V_{SS}
13	DQS0	14	DQ6	63	DQ25	64	DQ29	113	$\overline{CAS}$	114	ODT0	163	NC,TEST	164	CK1
15	V_{SS}	16	DQ7	65	V_{SS}	66	V_{SS}	115	NC/ $\overline{S1}$	116	NC / A13	165	V_{SS}	166	$\overline{CK1}$
17	DQ2	18	V_{SS}	67	DM3	68	$\overline{DQS3}$	117	V_{DD}	118	V_{DD}	167	$\overline{DQS6}$	168	V_{SS}
19	DQ3	20	DQ12	69	**NC/$\overline{RESET}$**	70	DQS3	119	NC / ODT1	120	**NC/$\overline{S3}$**	169	DQS6	170	DM6
21	V_{SS}	22	DQ13	71	V_{SS}	72	V_{SS}	121	V_{SS}	122	V_{SS}	171	V_{SS}	172	V_{SS}
23	DQ8	24	V_{SS}	73	DQ26	74	DQ30	123	DQ32	124	DQ36	173	DQ50	174	DQ54
25	DQ9	26	DM1	75	DQ27	76	DQ31	125	DQ33	126	DQ37	175	DQ51	176	DQ55
27	V_{SS}	28	V_{SS}	77	V_{SS}	78	V_{SS}	127	V_{SS}	128	V_{SS}	177	V_{SS}	178	V_{SS}
29	$\overline{DQS1}$	30	CK0	79	CKE0	80	NC / CKE1	129	$\overline{DQS4}$	130	DM4	179	DQ56	180	DQ60
31	DQS1	32	$\overline{CK0}$	81	V_{DD}	82	V_{DD}	131	DQS4	132	V_{SS}	181	DQ57	182	DQ61
33	V_{SS}	34	V_{SS}	83	**NC/$\overline{S2}$**	84	NC / A15	133	V_{SS}	134	DQ38	183	V_{SS}	184	V_{SS}
35	DQ10	36	DQ14	85	NC/BA2	86	NC / A14	135	DQ34	136	DQ39	185	DM7	186	$\overline{DQS7}$
37	DQ11	38	DQ15	87	V_{DD}	88	V_{DD}	137	DQ35	138	V_{SS}	187	V_{SS}	188	DQS7
39	V_{SS}	40	V_{SS}	89	A12	90	A11	139	V_{SS}	140	DQ44	189	DQ58	190	V_{SS}
41	V_{SS}	42	V_{SS}	91	A9	92	A7	141	DQ40	142	DQ45	191	DQ59	192	DQ62
43	DQ16	44	DQ20	93	A8	94	A6	143	DQ41	144	V_{SS}	193	V_{SS}	194	DQ63
45	DQ17	46	DQ21	95	V_{DD}	96	V_{DD}	145	V_{SS}	146	$\overline{DQS5}$	195	SDA	196	V_{SS}
47	V_{SS}	48	V_{SS}	97	A5	98	A4	147	DM5	148	DQS5	197	SCL	198	SA0
49	$\overline{DQS2}$	50	$\overline{Event}$	99	A3	100	A2	149	V_{SS}	150	V_{SS}	199	V_{DD}SPD	200	SA1

Note:
1. NC = No Connect; NC,TEST(pin 163) is for bus analysis tool and is not connected on normal memory modules.
2. Pins 69, 83, 120 used by 4 rank DDR2 SO-DIMMs.

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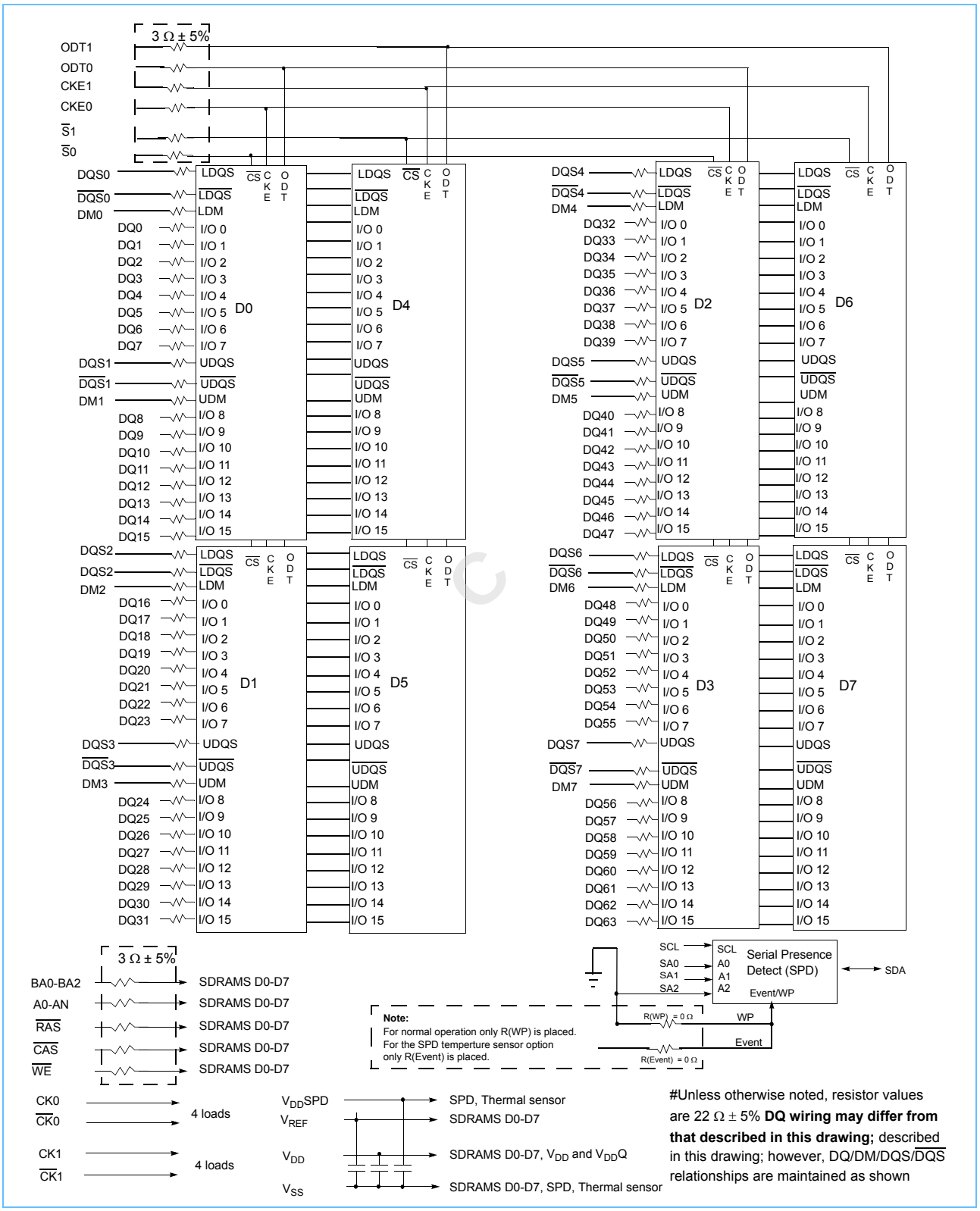
Block Diagram: Raw Card Version A, G (Populated as 2 ranks of x16 SDRAMs)



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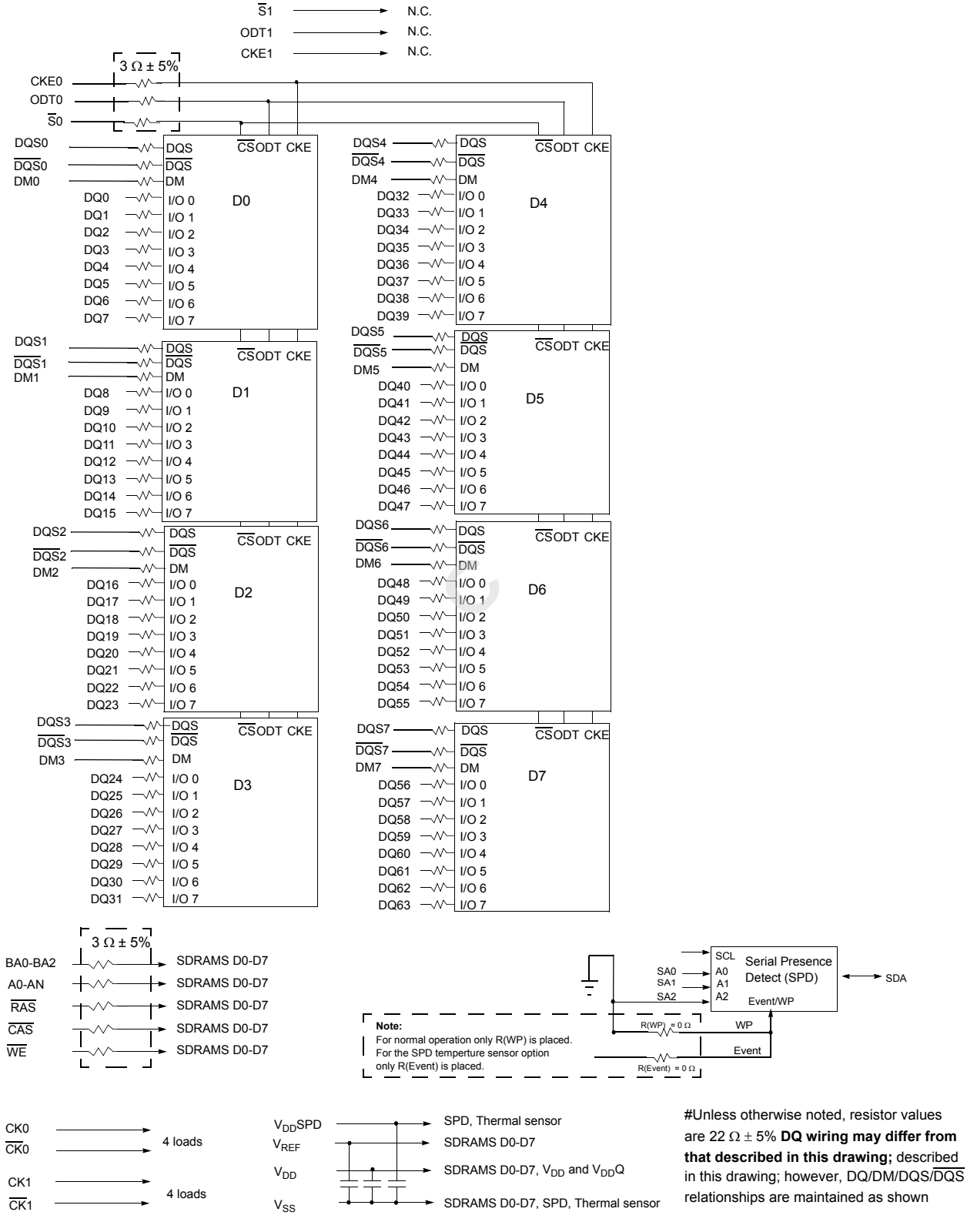
Block Diagram: Raw Card Version B(Populated as 1 rank of x8 SDRAMs)



#Unless otherwise noted, resistor values are 22 Ω ± 5% **DQ wiring may differ from that described in this drawing;** described in this drawing; however, DQ/DM/DQS/$\overline{\text{DQS}}$ relationships are maintained as shown

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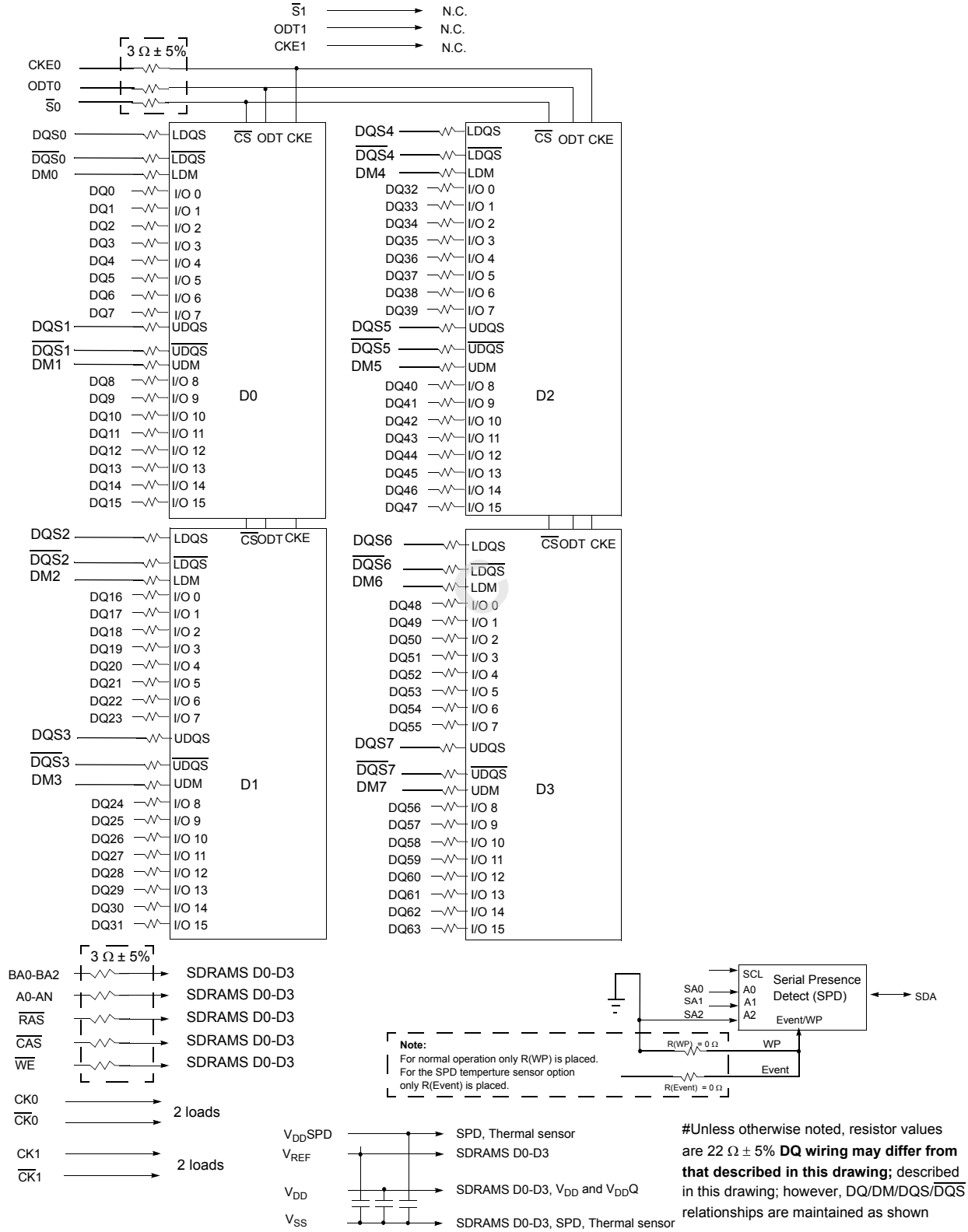
Block Diagram: Raw Card Version C, H (Populated as 1 rank of x16 SDRAMs)



#Unless otherwise noted, resistor values are 22 Ω ± 5% **DQ wiring may differ from that described in this drawing;** described in this drawing; however, DQ/DM/DQS/$\overline{DQS}$ relationships are maintained as shown

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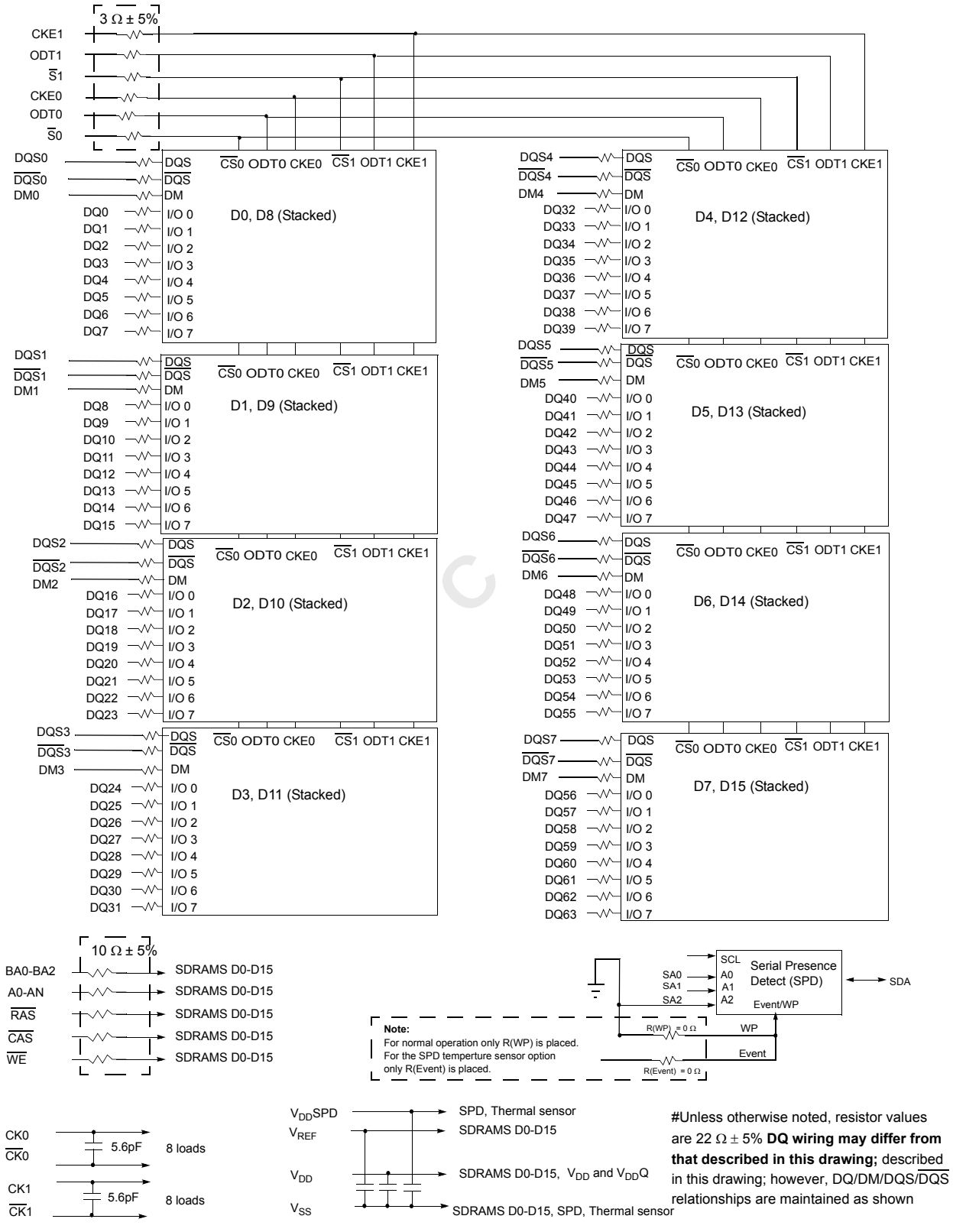
Block Diagram: Raw Card Version D (Populated as 2 rank of x8 SDRAMs)



#Unless otherwise noted, resistor values are 22 Ω ± 5% **DQ wiring may differ from that described in this drawing;** described in this drawing; however, DQ/DM/DQS/$\overline{\text{DQS}}$ relationships are maintained as shown

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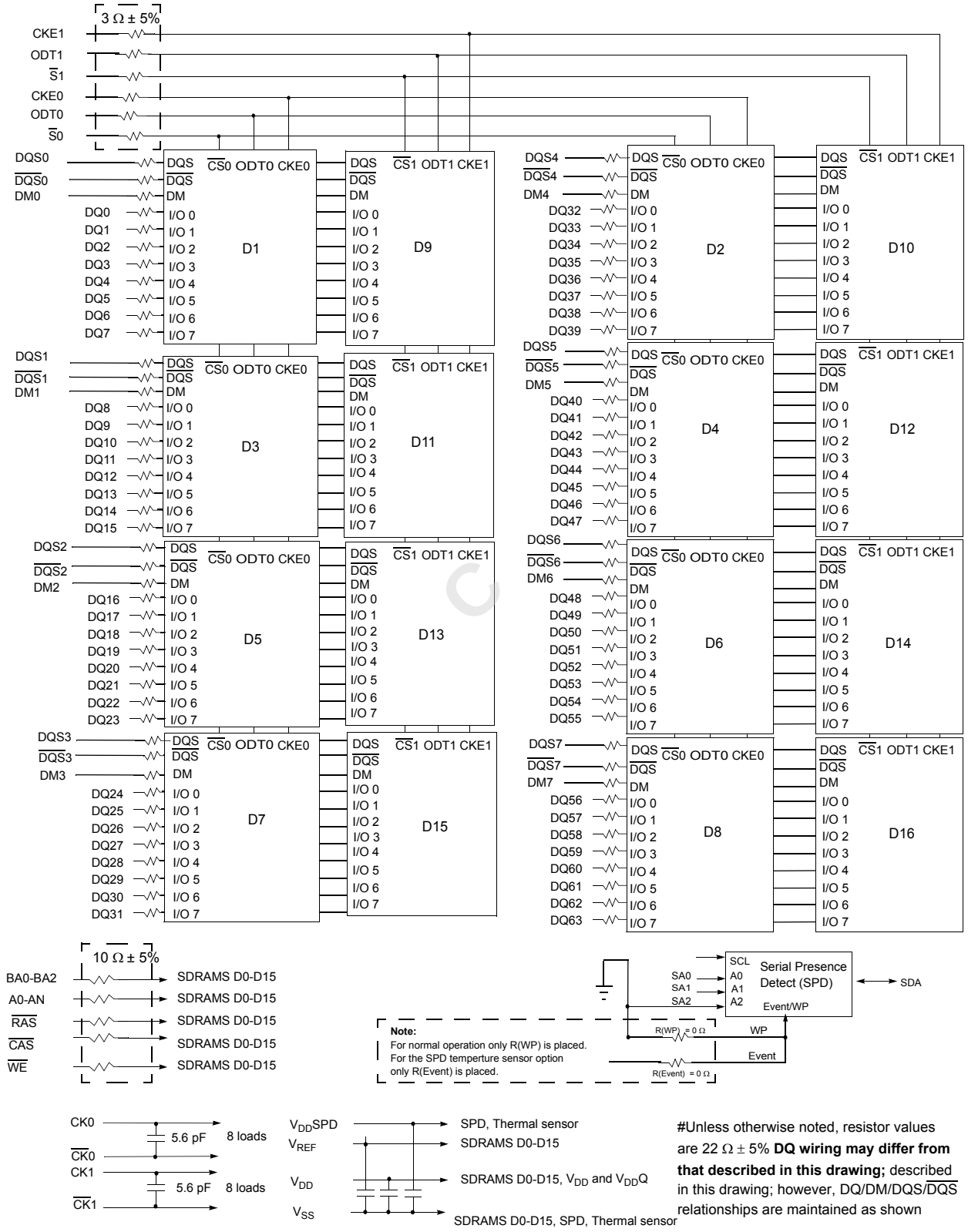
Block Diagram: Raw Card Version E (populated as 2 rank of x8 SDRAMs)



#Unless otherwise noted, resistor values are 22 Ω ± 5% **DQ wiring may differ from that described in this drawing;** described in this drawing; however, DQ/DM/DQS/$\overline{DQS}$ relationships are maintained as shown

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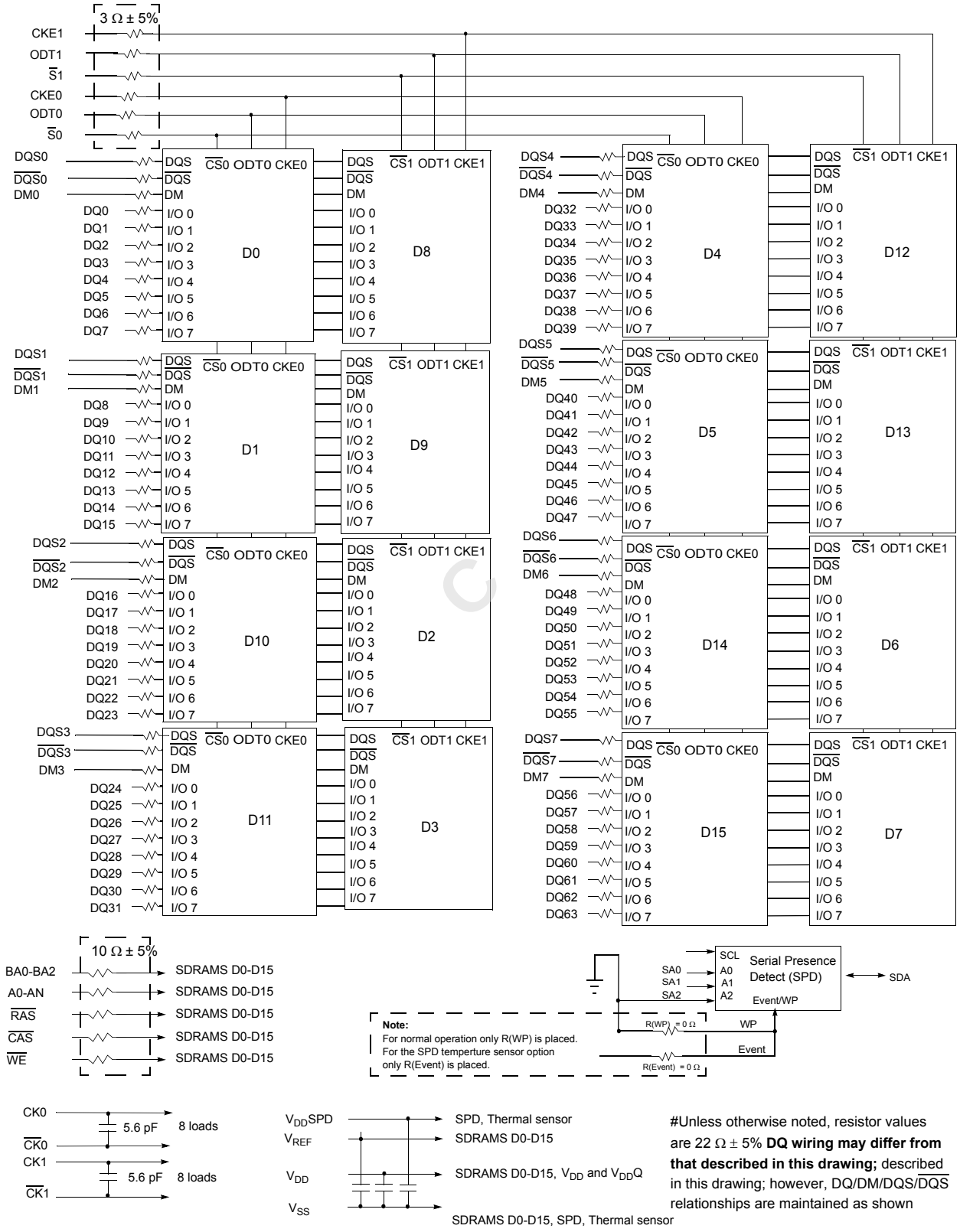
Block Diagram: Raw Card Version F (populated as 2 rank of x8 SDRAMs)



#Unless otherwise noted, resistor values are 22 Ω ± 5% **DQ wiring may differ from that described in this drawing;** described in this drawing; however, DQ/DM/DQS/$\overline{\text{DQS}}$ relationships are maintained as shown

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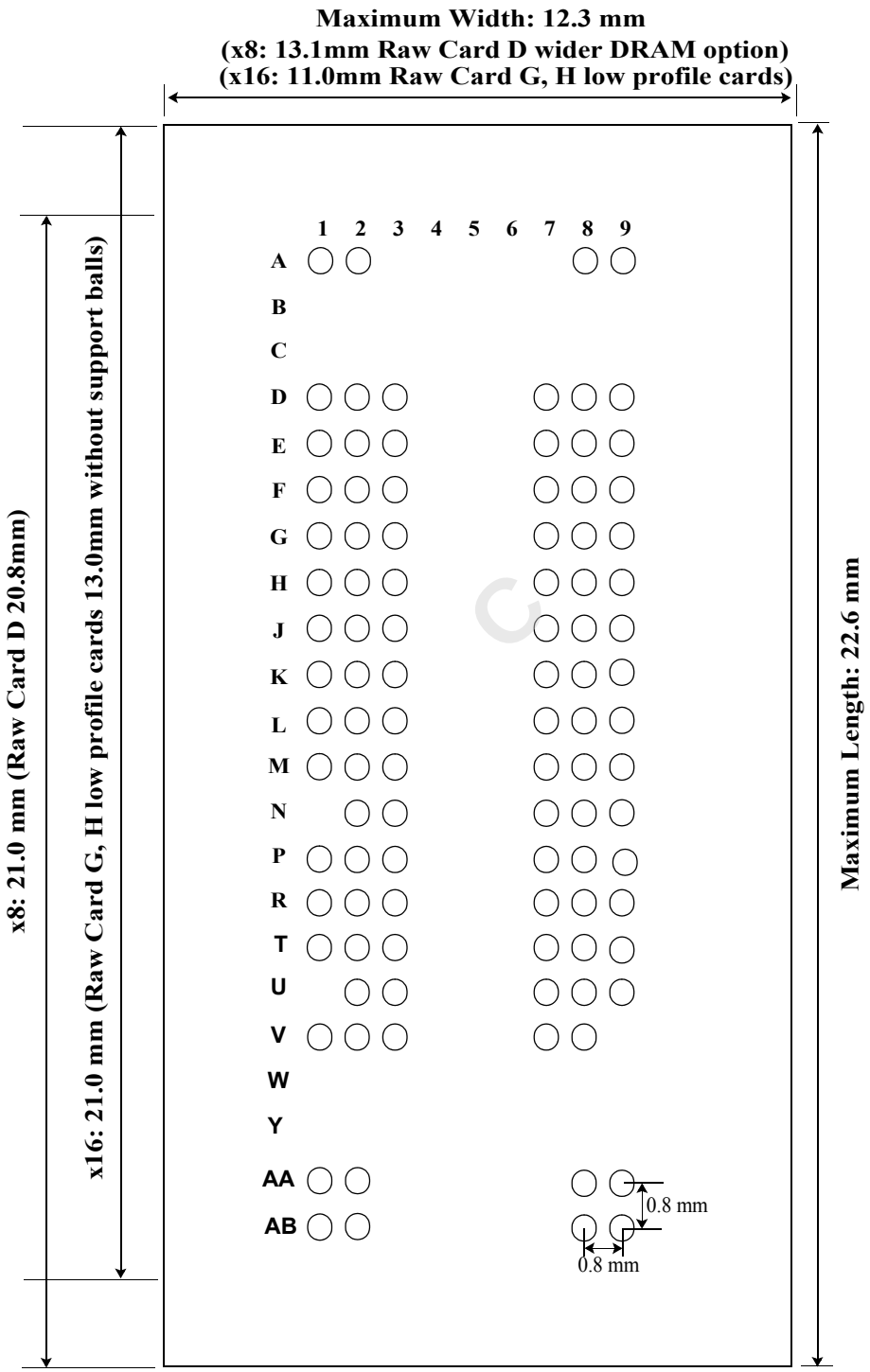
4. Component Details

SO-DIMM Common landing pattern for x8 and x16 -

256 Mb, 512 Mb, 1 Gb, 2 Gb, and 4 Gb DDR2 SDRAM Planar Components (Top View)



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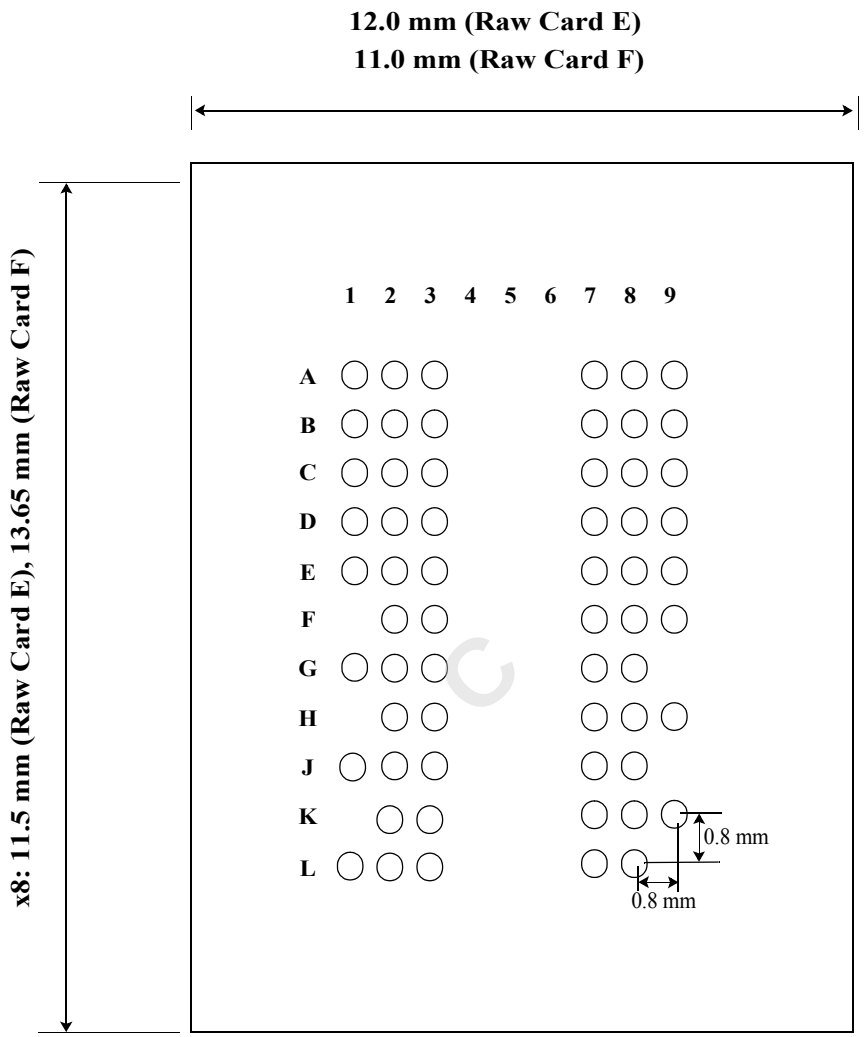

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x8 Ballout for 256 Mb, 512 Mb, 1 Gb, 2 Gb and 4 Gb DDR2 SDRAMs (Top View)

1	2	3		7	8	9
NC	NC		**A**		NC	NC
			B			
			C			
VDD	NC	VSS	**D**	VSSQ	NC	VDDQ
NC	NC	NC	**E**	NC	NC	NC
NC	NC	NC	**F**	NC	NC	NC
NC	NC	NC	**G**	NC	NC	NC
VDD	NU/$\overline{\text{RDQS}}$	VSS	**H**	VSSQ	$\overline{\text{DQS}}$	VDDQ
DQ6	VSSQ	DM/RDQS	**J**	DQS	VSSQ	DQ7
VDDQ	DQ1	VDDQ	**K**	VDDQ	DQ0	VDDQ
DQ4	VSSQ	DQ3	**L**	DQ2	VSSQ	DQ5
VDDL	VREF	VSS	**M**	VSSDL	CK	VDD
	CKE0	$\overline{\text{WE}}$	**N**	$\overline{\text{RAS}}$	$\overline{\text{CK}}$	ODT
BA2	BA0	BA1	**P**	$\overline{\text{CAS}}$	$\overline{\text{CS}}$0	$\overline{\text{CS}}$1
CKE1	A10	A1	**R**	A2	A0	VDD
VSS	A3	A5	**T**	A6	A4	ODT1
	A7	A9	**U**	A11	A8	VSS
VDD	A12	A14	**V**	A15	A13	
			W			
			Y			
NC	NC		**AA**		NC	NC
NC	NC		**AB**		NC	NC

1. The above landing pattern also applies for stacked components. For non-stacked components, remove pads at R1 (CKE1), P9($\overline{\text{CS}}$1) and T9 (ODT1) locations.

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x16 Ballout for 256 Mb, 512 Mb, 1 Gb, 2 Gb and 4 Gb DDR2 SDRAMs (Top View)

1	2	3		7	8	9
NC	NC		**A**		NC	NC
			B			
			C			
VDD	NC	VSS	**D**	VSSQ	$\overline{\text{UDQS}}$	VDDQ
UDQ6	VSSQ	UDM	**E**	UDQS	VSSQ	UDQ7
VDDQ	UDQ1	VDDQ	**F**	VDDQ	UDQ0	VDDQ
UDQ4	VSSQ	UDQ3	**G**	UDQ2	VSSQ	UDQ5
VDD	NC	VSS	**H**	VSSQ	$\overline{\text{LDQS}}$	VDDQ
LDQ6	VSSQ	LDM	**J**	LDQS	VSSQ	LDQ7
VDDQ	LDQ1	VDDQ	**K**	VDDQ	LDQ0	VDDQ
LDQ4	VSSQ	LDQ3	**L**	LDQ2	VSSQ	LDQ5
VDDL	VREF	VSS	**M**	VSSDL	CK	VDD
	CKE	$\overline{\text{WE}}$	**N**	$\overline{\text{RAS}}$	$\overline{\text{CK}}$	ODT
BA2	BA0	BA1	**P**	$\overline{\text{CAS}}$	$\overline{\text{CS}}$	
	A10	A1	**R**	A2	A0	VDD
VSS	A3	A5	**T**	A6	A4	
	A7	A9	**U**	A11	A8	VSS
VDD	A12	A14	**V**	A15	A13	
			W			
			Y			
NC	NC		**AA**		NC	NC
NC	NC		**AB**		NC	NC

DDR2 SDRAM FBGA Component Specifications

The DDR2 SDRAM components used with this DIMM design specification are intended to be consistent with JEDEC MO-207, variations DL-Z, DK-Z, DM-Z, DJ-Z. The MO-207 specification allows for a maximum component width of 12.5 mm. The DDR2 SO-DIMM is designed for a varitiy of maximum component widths and maximum lengths, refer to the applicable raw card for exact componet size allowed. Components used on DDR2 SO-DIMMs are also limited to a maximum height (as shown in dimension "A" of MO-207) of 1.35 mm.

Reference SPD Component Specifications

The Serial Presence Detect EEPROMs have their own power pin, V_{DD}SPD, so that they can be programmed or read without powering up the rest of the module. The wide voltage range permits use with 1.8V, 2.5V or 3.3V serial buses.

DC Electrical Characteristics

Symbol	Parameter	Min	Max	Units	Notes
V_{DD}SPD	Core Supply Voltage	1.7	3.6	V	1

Note: Note: SO-DIMMs that include an optional temperature sensor may require a restricted VDDSPD operating voltage range for proper operation of the temperature sensor. Refer to the thermal sensor specification for details regarding the supported voltage range. All other functions of the SO-DIMM SPD are supported across the full VDDSPD range.

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5. Unbuffered SO-DIMM Details

DDR2 SDRAM Module Configurations (Reference Designs)

Raw Card	SO-DIMM Capacity	SO-DIMM Organization	SDRAM Density	SDRAM Organization	# of SDRAMs	# of Ranks	SDRAM Package Type	# of banks in SDRAM	# Address bits row/col
A, G	256 MB	32 M x 64	256 Mbit	16 M x 16	8	2	FBGA	4	13/9
A, G	512 MB	64 M x 64	512 Mbit	32 M x 16	8	2	FBGA	4	13/10
A, G	1 GB	128 M x 64	1 Gbit	64 M x 16	8	2	FBGA	8	13/10
A, G	2 GB	256 M x 64	2 Gbit	128 M x 16	8	2	FBGA	8	14/10
A, G	4 GB	512 M x 64	4 Gbit	256 M x 16	8	2	FBGA	8	15/10

Raw Card	SO-DIMM Capacity	SO-DIMM Organization	SDRAM Density	SDRAM Organization	# of SDRAMs	# of Ranks	SDRAM Package Type	# of banks in SDRAM	# Address bits row/col
B	256 MB	32 M x 64	256 Mbit	32 M x 8	8	1	FBGA	4	13/10
B	512 MB	64 M x 64	512 Mbit	64 M x 8	8	1	FBGA	4	14/10
B	1 GB	128 M x 64	1 Gbit	128 M x 8	8	1	FBGA	8	14/10
B	2 GB	256 M x 64	2 Gbit	256 M x 8	8	1	FBGA	8	15/10
B	4 GB	512 M x 64	4 Gbit	512 M x 8	8	1	FBGA	8	16/10

Raw Card	SO-DIMM Capacity	SO-DIMM Organization	SDRAM Density	SDRAM Organization	# of SDRAMs	# of Ranks	SDRAM Package Type	# of banks in SDRAM	# Address bits row/col
C, H	128 MB	16 M x 64	256 Mbit	16 M x 16	4	1	FBGA	4	13/9
C, H	256 MB	32 M x 64	512 Mbit	32 M x 16	4	1	FBGA	4	13/10
C, H	512 MB	64 M x 64	1 Gbit	64 M x 16	4	1	FBGA	8	13/10
C, H	1 GB	128 M x 64	2 Gbit	128 M x 16	4	1	FBGA	8	14/10
C, H	2 GB	256 M x 64	4 Gbit	256 M x 16	4	1	FBGA	8	15/10

Raw Card	SO-DIMM Capacity	SO-DIMM Organization	SDRAM Density	SDRAM Organization	# of SDRAMs	# of Ranks	SDRAM Package Type	# of banks in SDRAM	# Address bits row/col
D	512 MB	64 M x 64	256 Mbit	32 M x 8	16	2	FBGA	4	13/10
D	1 GB	128 M x 64	512 Mbit	64 M x 8	16	2	FBGA	4	14/10
D	2 GB	256 M x 64	1 Gbit	128 M x 8	16	2	FBGA	8	14/10
D	4 GB	512 M x 64	2 Gbit	256 M x 8	16	2	FBGA	8	15/10
D	8 GB	1024 M x 64	4 Gbit	512 M x 8	16	2	FBGA	8	16/10

1. Raw Card D is a stacked solution.

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Raw Card	SO-DIMM Capacity	SO-DIMM Organization	SDRAM Density	SDRAM Organization	# of SDRAMs	# of Ranks	SDRAM Package Type	# of banks in SDRAM	# Address bits row/col
E, F	512 MB	64 M x 64	256 Mbit	32 M x 8	16	2	FBGA	4	14/10
E, F	1 GB	128 M x 64	512 Mbit	64 M x 8	16	2	FBGA	4	14/10
E, F	2 GB	256 M x 64	1 Gbit	128 M x 8	16	2	FBGA	8	14/10
E, F	4 GB	512 M x 64	2 Gbit	256 M x 8	16	2	FBGA	8	15/10

1. Raw Card E and F are planar solutions.

Input Loading Matrix

Signal Names	Input Device	R/C A, G	R/C B	R/C C, H	R/C D	R/C E	R/C F
Clock (CK0, $\overline{CK}$0 - CK1, $\overline{CK}$1)	SDRAM	4	4	2	8	8	8
CKE0/CKE1/$\overline{S}$0/$\overline{S}$1/ODT0/ODT1	SDRAM	4	8	4	8	8	8
Addr/$\overline{RAS}$/$\overline{CAS}$/BA/$\overline{WE}$	SDRAM	8	8	4	16	16	16
DQ/DQS/$\overline{DQS}$/DM	SDRAM	2	1	1	2	2	2
SCL/SDA/SA	EEPROM	1	1	1	1	1	1

1. Raw card D is a stacked solution, raw card E and F are planar solutions.

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DDR2 SO-DIMM Gerber File Releases

Reference design file updates will be released as needed. This specification will reflect the most recent design files, but may be updated to reflect clarifications to the specification only; in these cases, the design files will not be updated. The following table outlines the most recent design file releases

Note: Future design file releases will include both a date and a revision label. All changes to the design file are also documented within the 'read-me' file.

Raw Card	Specification Revision	Applicable Design File	Notes
A	2.1	PC2-6400_SODIMM_V210_RC_A0_20041027	
B	2.1	PC2-6400_SO-DIMM_V210_RC_B0_20041214	
C	2.1	PC2-6400_SODIMM_V210_RC_C0_20041027	
D	2.1	PC2-6400_SO-DIMM_V210_RC_D0_20041029	
E	2.3	PC2-6400_SODIMM_V23_RC_E1_20070426	
F	2.4	PC2-6400_SODIMM_V24_RC_F1_20070711	
G	2.5		
H	2.5		

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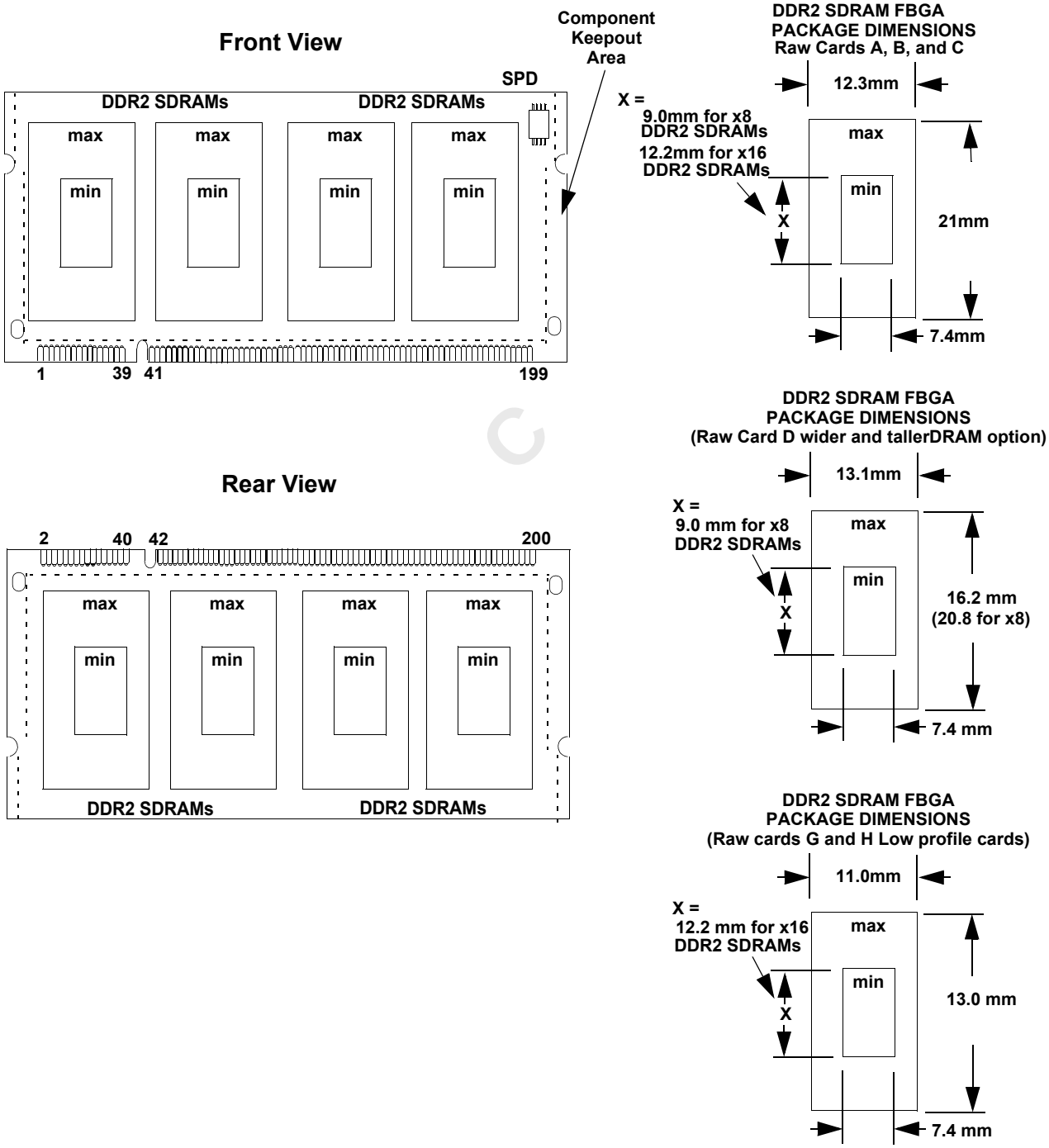
Example Raw Card Component Placement

Raw Cards A, B, C, D, G, H

The component layout for Raw Cards A, B, C, D, G and H are similar. In the case of Raw Card C and H, DDR2 SDRAMs will be included on the front side of the card; however, passive components are on both sides of the board. This example is for reference only; refer to JEDEC standard MO-224 variation AB/CB for details.



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Raw Cards E and F

The component layout for raw cards E and F are similar. In the case of raw card E the DRAMs are placed vertical and for raw card F the DRAMs are placed horizontal to the edge connector. This example is for reference only; refer to JEDEC standard MO-224 for details

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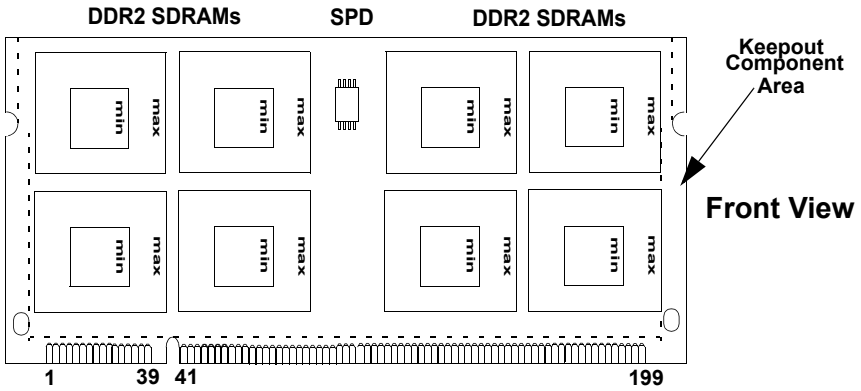

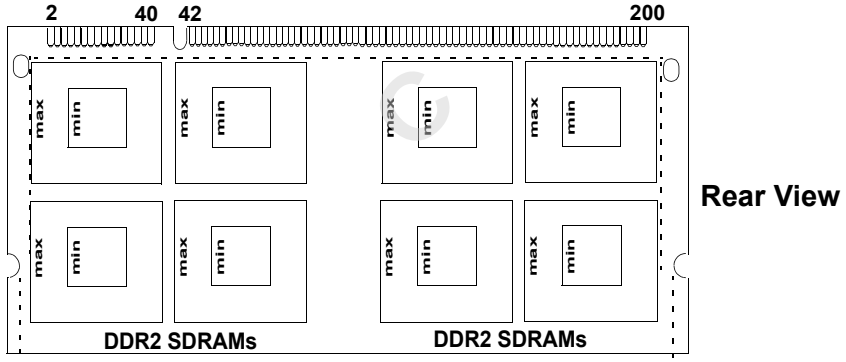

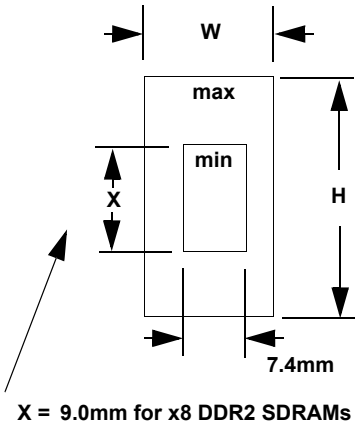

PACKAGE DIMENSIONS DDR2 SDRAM FBGA

Raw Cards E: W = 12.0, H = 11.5

Raw Cards F: W = 11.0, H = 13.65

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6. SO-DIMM Wiring Details

Signal Groups

This specification categorizes SDRAM timing-critical signals into four groups whose members have identical loadings and routings. The following table summarizes the signals contained in each group.

Signal Group	Signals In Group	Page
Clocks for Unbuffered SO-DIMM	CK [1:0], $\overline{\text{CK}}$ [1:0]	25-29
Data, Data Mask, Data Strobe	DQ [63:0], DM[7:0], DQS[7:0], $\overline{\text{DQS}}$[7:0]	30-31
Select, Clock Enable, ODT	$\overline{\text{S}}$ [1:0], CKE [1:0], ODT[1:0]	32-36
Address/Command	Ax, BAx, $\overline{\text{RAS}}$, $\overline{\text{CAS}}$, $\overline{\text{WE}}$, ODT[1:0]	37-41

General Net Structure Routing Guidelines

Net structures and lengths must satisfy signal quality and setup/hold time requirements for the memory interface. Net structure diagrams for each signal group are shown in the following sections. Each diagram is accompanied by a trace length table that lists the minimum and maximum allowable lengths for each trace segment and/or net.

The general routing recommendations are as follows. Other stackups and layouts are possible that meet the electrical characteristics.

- Route all signal traces except clocks using 4 mil rules, 4 mil minimum spacing between adjacent traces.
- Route clocks as much as possible using the inner layers.
- Test points are required on all signal groups.

Explanation of Net Structure Diagrams

The net structure routing diagrams provide a reference design example for each raw card version. These designs provide an initial basis for unbuffered SO-DIMM designs. The diagrams should be used to determine individual signal wiring on a SO-DIMM for any supported configuration. Only transmission lines (represented as cylinders and labeled with trace length designators “TL”) represent physical trace segments. All other lines are zero in length. To verify SO-DIMM functionality, a full simulation of all signal integrity and timing is required. The given net structures and trace lengths are not inclusive for all solutions.

Once the net structure has been determined, the permitted trace lengths for the net structure can be read from the table below each net structure routing diagram. Some configurations require the use of multiple net structure routing diagrams to account for varying load quantities on the same signal. All diagrams define one load as one DDR2 SDRAM input unless specified as two loads (stacked DRAM has two loads). It is highly recommended that the net structure routing data in this document be simulated by the user.

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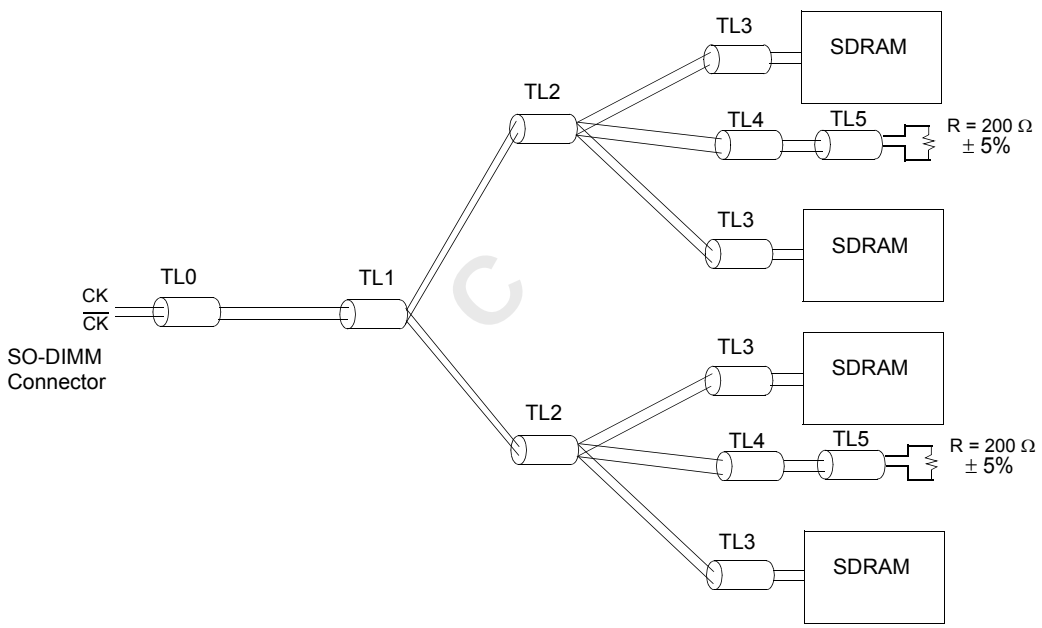
Differential Clock Net Structures

CK[1:0], $\overline{CK}$[1:0]

DDR2 SDRAM clock signals must be carefully routed to meet the following requirements:

- Signal quality
- Rise/Fall time
- Cross point of the differential pair in the SDRAM
- JEDEC-compatible reference delays
- Minimal segment length differences (less than 2.54 mm total) between clocks of the same function

Clock Net Wiring CK[1:0], $\overline{CK}$[1:0] (Raw Cards A, B and G)



Clock Routing Trace Lengths

Raw card	TL0 Outer		TL1 Inner		TL2 Inner		TL3 Outer		TL4 Inner		TL5 Outer		Notes
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A, G	4.8	5.0	9.4	9.6	9.7	9.9	2.9	3.1	6.8	7.0	0.7	0.9	1, 2
B	2.8	2.9	12.2	12.4	8.6	8.8	2.9	3.1	6.5	6.7	0.7	0.9	1, 2

1. All distances are given in millimeters and must be kept within a tolerance of ± 0.8 millimeters.
2. Total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN} = TL0_{MIN} + TL1_{MIN} + TL2_{MIN} + TL3_{MIN}$
   Total $Length_{MAX} = TL0_{MAX} + TL1_{MAX} + TL2_{MAX} + TL3_{MAX}$

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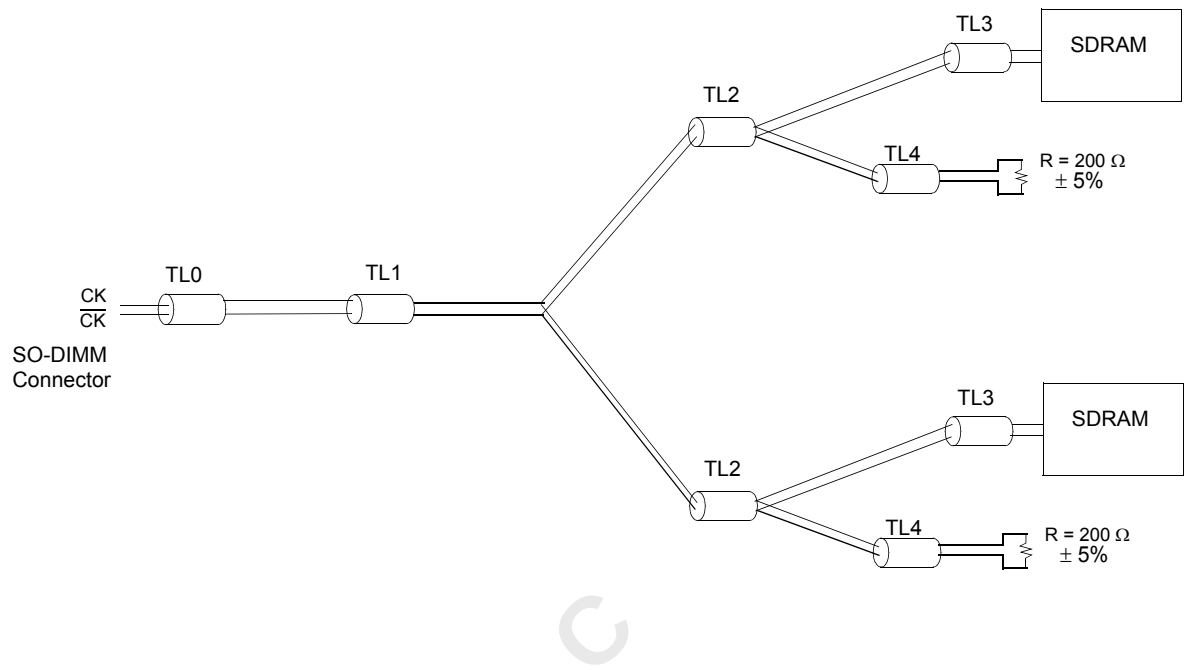
Clock Net Wiring CK[1:0], $\overline{CK}$[1:0] (Raw Card C and H)



Clock Routing Trace Lengths

Raw card	TL0 Outer		TL1 Inner		TL2 Inner		TL3 Outer		TL4 Outer		Notes
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
C, H	4.8	5.0	27.4	27.6	7.6	7.9	2.9	3.1	2.4	2.6	1, 2

1. All distances are given in millimeters and must be kept within a tolerance of ± 0.8 millimeters
2. Total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN} = TL0_{MIN} + TL1_{MIN} + TL2_{MIN} + TL3_{MIN}$
   Total $Length_{MAX} = TL0_{MAX} + TL1_{MAX} + TL2_{MAX} + TL3_{MAX}$

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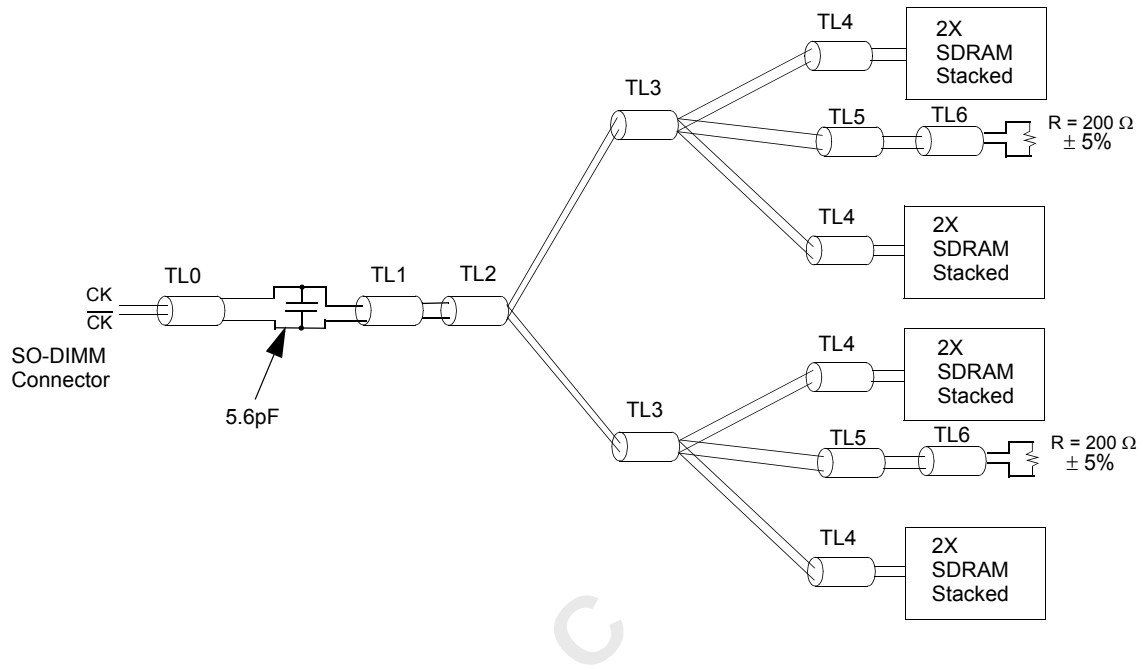
Clock Net Wiring CK[1:0], $\overline{CK}$[1:0] (Raw Card D)



Clock Routing Trace Lengths

Raw card	TL0 Outer		TL1 Outer		TL2 Inner		TL3 Inner		TL4 Outer		TL5 Inner		TL6 Outer		Notes
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
D	3.3	3.5	0.8	1.0	10.9	11.1	8.6	8.8	4.2	4.3	6.5	6.7	0.7	2.7	1, 2

1. All distances are given in millimeters and must be kept within a tolerance of ± 0.8 millimeters
2. Total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN} = TL0_{MIN} + TL1_{MIN} + TL2_{MIN} + TL3_{MIN} + TL4_{MIN}$
   Total $Length_{MAX} = TL0_{MAX} + TL1_{MAX} + TL2_{MAX} + TL3_{MAX} + TL4_{MAX}$

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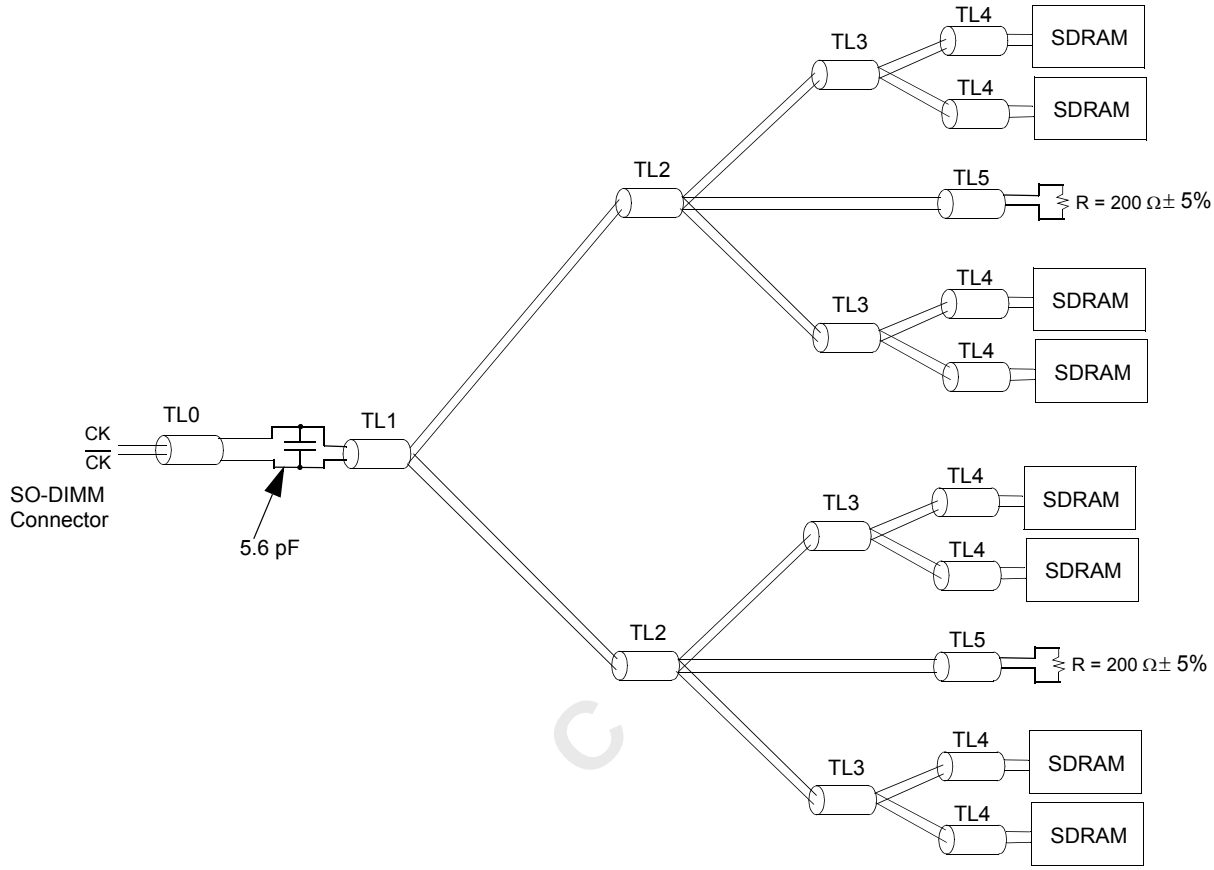
Clock Net Wiring CK[1:0], $\overline{CK}$[1:0] (Raw Card E)



Clock Routing Trace Lengths

Raw card	TL0 Outer		TL1 Outer		TL2 Inner		TL3 Inner		TL4 Outer		TL5 Outer		Notes
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
E	3.2	3.3	9.4	9.5	12.0	12.1	8.3	8.4	4.0	4.1	0.6	0.7	1, 2,3

1. All distances are given in millimeters and must be kept within a tolerance of ± 0.8 millimeters
2. Total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN} = TL0_{MIN} + TL1_{MIN} + TL2_{MIN} + TL3_{MIN} + TL4_{MIN}$
   Total $Length_{MAX} = TL0_{MAX} + TL1_{MAX} + TL2_{MAX} + TL3_{MAX} + TL4_{MAX}$
3. Clock must be routed with 4/6/4 mil trace width/spacing/width.

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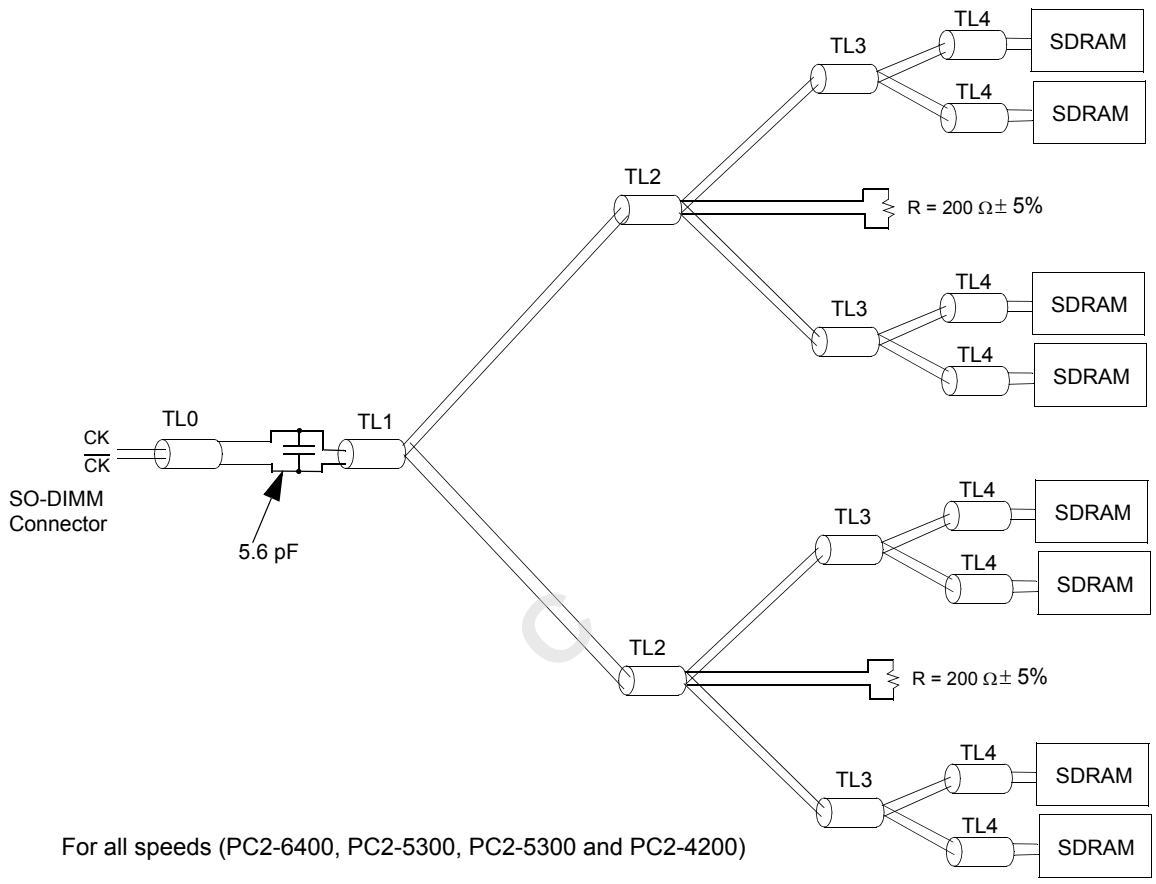
Clock Net Wiring CK[1:0], $\overline{CK}$[1:0] (Raw Card F)



Clock Routing Trace Lengths

Raw card	TL0 Inner		TL1 Inner		TL2 Inner		TL3 Inner		TL4 Outer		Notes
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
F	4.5	4.5	12.3	12.3	10.3	10.3	7.2	7.2	3.7	3.7	1,2,3

1. All distances are given in millimeters and must be kept within a tolerance of ± 0.8 millimeters
2. Total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total Length_{MIN} = TL0 + TL1 + TL2 + TL3 + TL4 = 38.0 mm
   Total Length_{MAX} = TL0 + TL1 + TL2 + TL3 + TL4 = 38.0 mm
3. Clock must be routed with 4/6/4 mil trace width/spacing/width.

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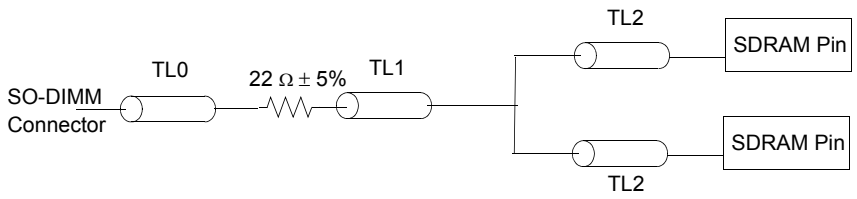
Data Net Structures

DQ[63:0], DM[7:0], DQS[7:0], $\overline{\text{DQS}}$[7:0]

Special attention has been paid to balancing the data nets within a DDR2 SDRAM, within a particular SO-DIMM, and across the SO-DIMM family. Data nets have been placed in order to bound the data strobe nets. Because data travels with the data strobe, the placement of the strobe in the middle of the narrow window aids in data timing. Although it is not necessary to ensure consistent delays between SDRAMs and/or card types, doing so facilitates system design, system simulation, and DIMM specifications. It is recommend to maintain consistent delays for all nets as described in the following tables.

Net Structure Routing for DQ[63:0], DM[7:0], DQS[7:0], $\overline{\text{DQS}}$[7:0] (Raw card A, E, F, G)



Trace Lengths for Data Net Structure

Raw card	TL0 Outer		TL1 Inner		TL1 Outer		TL0 + TL1 (Inner / Outer)Total		TL2 Outer		Total		R1 Ohms	Notes
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
A, G	3.0	3.6	NA	NA	14.3	14.5	17.6	17.8	2.9	3.1	20.6	20.8	22	1,2,3,4
E	3.1	4.4	17.2	21.7	0.6	2.3	22.9	25.5	0.6	2.8	26.0	26.2	22	1,2,3,4
F (DQ)	3.0	7.2	23.7	27.9	26.2	30.4	33.4	33.4	0.6	0.6	34.0	34.0	22	1,3,4,5
F (DQS)	3.0	7.2	24.2	28.4	21.7	25.9	31.4	31.4	2.6	2.6	34.0	34.0	22	1,3,4,5

1. All distances are given in millimeters and must be kept within a tolerance of ± 0.8 millimeters (Raw card F has a tolerance of +/- 0.1mm)
2. Total Min and Total Max refer to the min and max respectively of TL0 + TL1 + TL2
3. TL0 and TL1 are given for lines without any via. For example, if there is a via on the the transmission line, then the line length has to be reduced by 2.54mm, and if there are two vias on the TL, then the line length has to be reduced by 5.08mm
4. These signals must be referenced to ground
5. For Raw Card A and E, TL1 is an outer layer. For Raw Card F, TL1 may be on the inner or outer layer.

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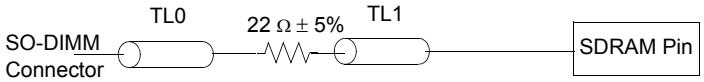
Data Net Structures DQ[63:0], DM[7:0], DQS[7:0], $\overline{\text{DQS}}$[7:0] (Raw cards B, C, D and H)

Net Structure Routing for Data, Data Mask, Data Strobe



Note: 2 SDRAM per package for raw card D

Trace Lengths for Data Net Structure

Raw card	TL0 Outer		TL1 Outer		Total		R1 Ohms	Notes
	Min	Max	Min	Max	Min	Max		
C, H	3.0	3.6	17.1	17.7	20.6	20.8	22	1,2,3,4
B	3.1	6.8	16.4	20.1	23.1	23.3	22	1,2,3,4
D	3.1	6.8	16.4	20.1	23.1	23.3	22	1,2,3,4

1. All distances are given in millimeters and must be kept within a tolerance of ± 0.8 millimeters
2. Total Min and Total Max refer to the min and max respectively of TL0 + TL1
3. TL0 and TL1 are given for lines without any via. For example, if there is a via on the the transmission line, then the line length has to be reduced by 2.54mm, and if there are two vias on the TL, then the line length has to be reduced by 5.08mm.
4. These signals must be referenced to ground

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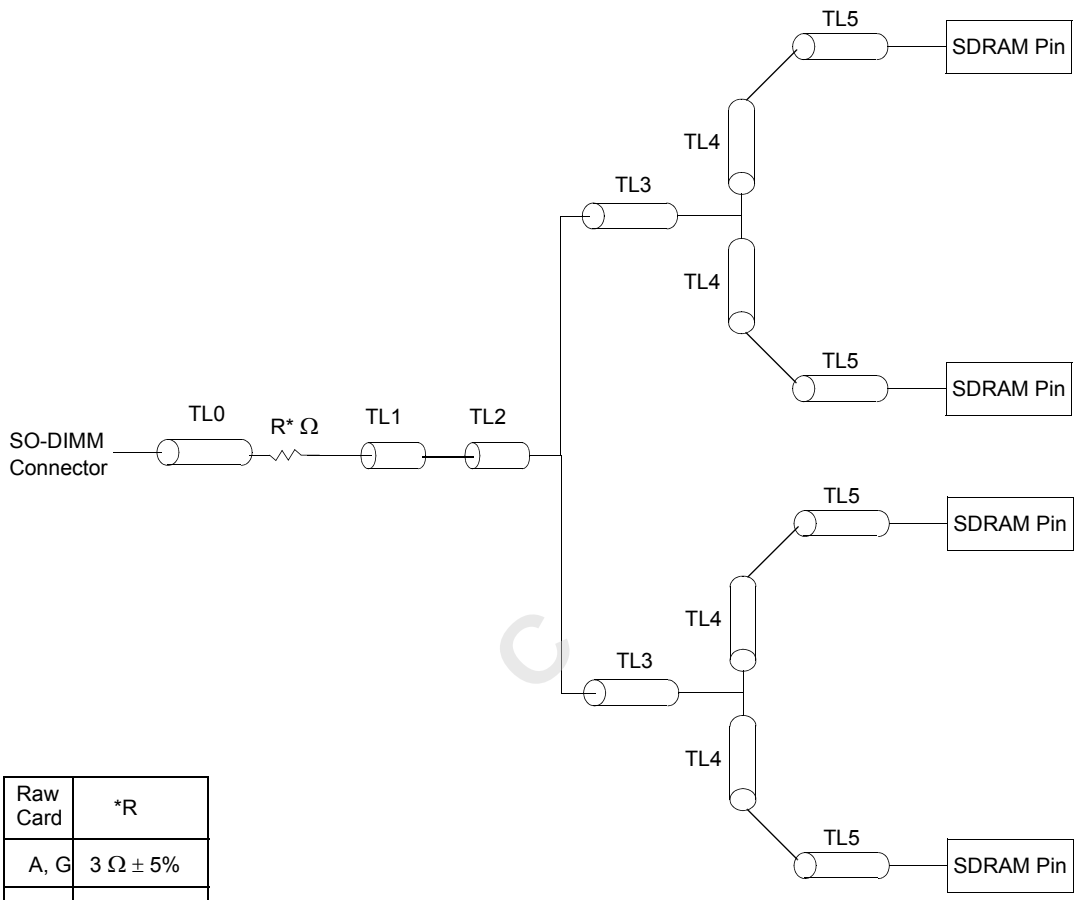
Control Net Structures $\overline{S}$[1:0], CKE[1:0], ODT[1:0] (Raw cards A, C, G and H)

Net Structure Routing for Control nets



Raw Card	*R
A, G	3 Ω ± 5%
C, H	3 Ω ± 5%

Trace Lengths for Control Net Structures

	TL0 Outer		TL1 Outer		TL2 Inner		TL1+TL2		TL3 Inner		TL4 Inner		TL5 Outer		Notes
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A, G	3.0	3.6	0.5	2.0	20.0	22.0	22.4	22.6	19.3	19.7	7.4	8.0	1.8	2.2	1,2
C, H	3.0	3.6	3.0	5.5	53.0	57.0	58.9	59.1	18.2	18.5	7.2	7.5	2.0	3.5	1,3

1. All distances are given in mm and should be kept within a tolerance of ± 0.8 mm.
2. For Raw Card A, the total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total Length_{MIN} = TL0 +TL1 + TL2 + $TL3_{MIN}$ +$TL4_{MIN}$ + $TL5_{MIN}$ = 54.7 *(gerber measured mininmun)*
   Total Length_{MAX} = TL0 + TL1 + TL2 + $TL3_{MAX}$ + $TL4_{MAX}$ + $TL5_{MAX}$ = 56.3 *(gerber measured maxinum)*
3. For Raw Card C, the total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total Length_{MIN} = TL0 + TL1 + TL2 + $TL3_{MIN}$ + $TL4_{MIN}$ + $TL5_{MIN}$ = 89.9 *(gerber measured mininmun)*
   Total Length_{MAX} = TL0 + TL1 + TL2 + $TL3_{MAX}$ + $TL4_{MAX}$ + $TL5_{MAX}$ = 91.4 *(gerber measured maxinum)*

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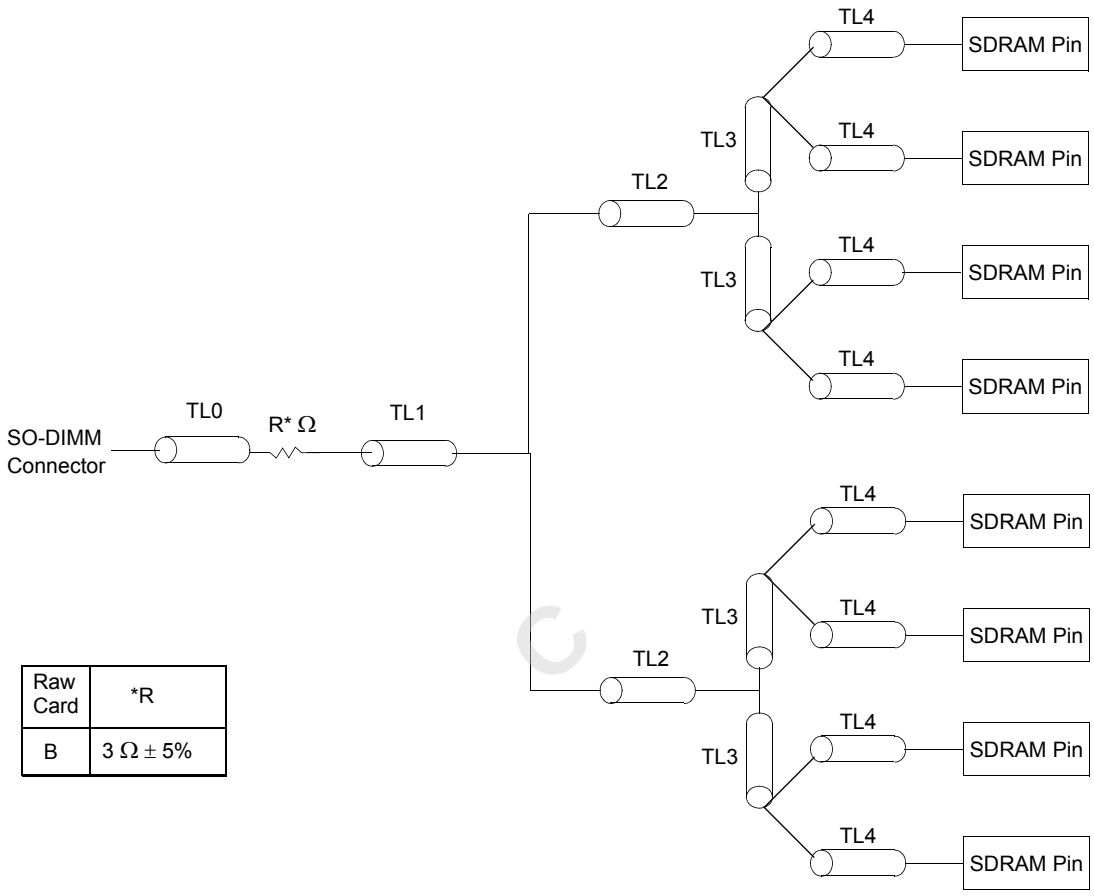
Control Net Structures $\overline{S}$[0], CKE[0], ODT[0] (Raw card B)

Net Structure Routing for Control Nets



Raw Card	*R
B	3 Ω ± 5%

Trace Lengths for Control Net Structure

	TL0 Outer		TL1 Outer		TL2 Inner		TL3 Inner		TL4 Outer	
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
B	3.5	3.7	28.8	29.1	19.2	19.4	7.9	7.9	3.5	4.0

1. All distances are given in mm and should be kept within a tolerance of ± 0.8 mm
2. The total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total Length_{MIN} = TL0 + TL1 + $TL2_{MIN}$ + $TL3_{MIN}$ + $TL4_{MIN}$ = 63.1 *(gerber measured mininmun)*
   Total Length_{MAX} = TL0 + TL1 + $TL2_{MAX}$ + $TL3_{MAX}$ + $TL4_{MAX}$ = 63.5 *(gerber measured maxinum)*

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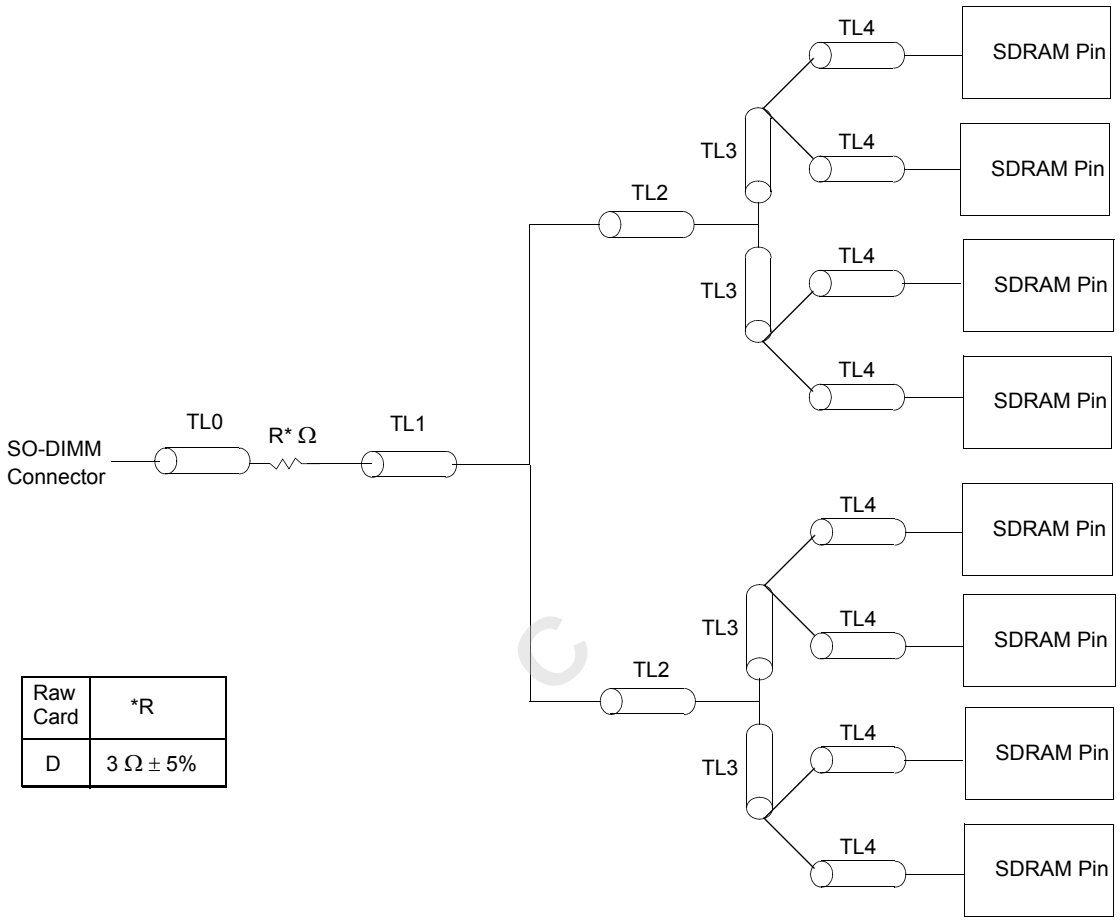
Control Net Structures $\overline{S}$[1:0], CKE[1:0], ODT[1:0] (Raw card D)

Net Structure Routing for Control Nets



Raw Card	*R
D	3 Ω ± 5%

Trace Lengths for Control Net Structure

	TL0 Outer		TL1 Outer		TL2 Inner		TL3 Inner		TL4 Outer	
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
D	3.4	4.2	25.9	29.0	17.9	19.2	7.9	7.9	3.3	10.8

1. All distances are given in mm and should be kept within a tolerance of ± 0.8 mm
2. The total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total Length_{MIN} = TL0 + TL1 + $TL2_{MIN}$ + $TL3_{MIN}$ + $TL4_{MIN}$ = 63.0 *(gerber measured mininmun)*
   Total Length_{MAX} = TL0 + TL1 + $TL2_{MAX}$ + $TL3_{MAX}$ + $TL4_{MAX}$ = 67.2 *(gerber measured maxinum)*

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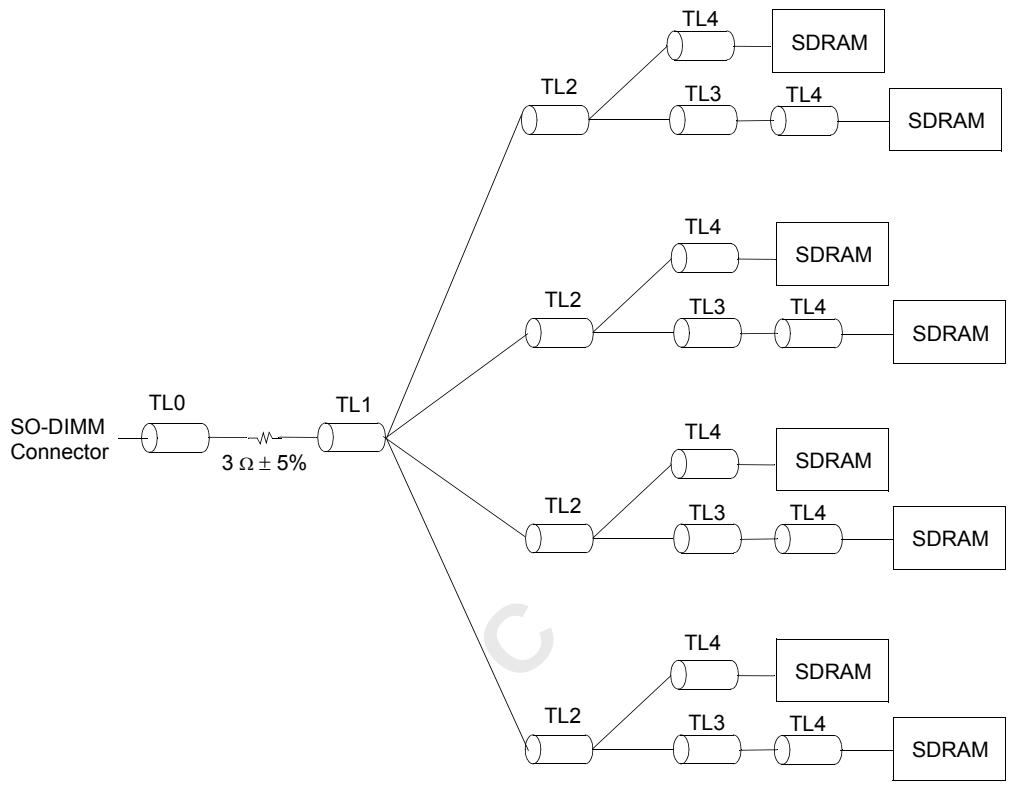
Control Net Structures $\overline{S}$[1:0], CKE[1:0], ODT[1:0] (Raw card E)

Net Structure Routing for Control Nets



Trace Lengths for Control Net Structure

	TL0 Outer		TL1 Outer		TL2 Inner		TL3 Inner		TL4 Outer		Note
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
E	3.4	4.1	22.1	22.8	14.1	19.8	14.0	14.2	1.8	3.6	1,2

1. All distances are given in mm and should be kept within a tolerance of ± 0.8 mm
2. The total routed trace lengths from card edge to DRAM pin are defined by the following equations:

   Total Length_{MIN} = TL0 + TL1 + $TL2_{MIN}$ + $TL4_{MIN}$ = 43.9 *(gerber measured mininmun)*

   Total Length_{MAX} = TL0 + TL1 + $TL2_{MAX}$ + $TL3_{MIN}$ + $TL4_{MAX}$ = 64.0 *(gerber measured maxinum)*

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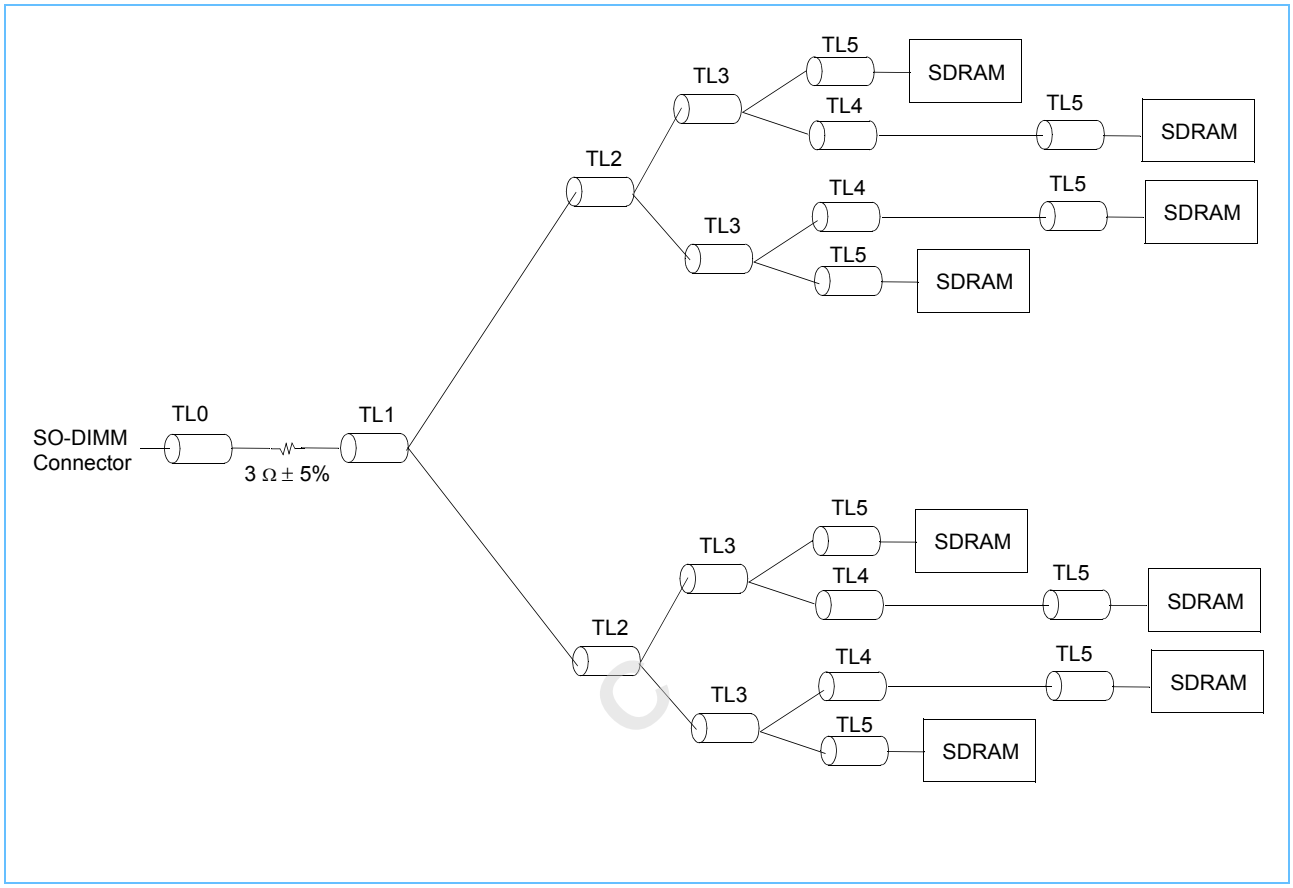
Control Net Structures $\overline{S}$[1:0], CKE[1:0], ODT[1:0] (Raw card F)

Net Structure Routing for Control Nets



Trace Lengths for Control Net Structure

	TL0 Outer		TL1 Inner /Outer		TL2 Inner		TL3 Inner		TL4 Inner		TL5 Outer		Note
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
F	3.2	5.9	10.6	13.3	19.5	19.5	11.3	11.3	13.0	13.3	3.7	3.7	1, 2, 3, 4

1. All distances are given in mm and should be kept within a tolerance of ± 0.1 mm
2. Total of TL0 + TL1 = 16.5mm
3. TL0 and TL1 are given for lines without any via. For example, if there is a via on the the transmission line, then the line length has to be reduced by 2.54mm, and if there are two vias on the TL, then the line length has to be reduced by 5.08mm.
4. The total routed trace lengths from card edge to DRAM pin are defined by the following equations:

   Total Length_{MIN} = (TL0 + TL1) + TL2 + TL3+ TL5_{MIN} = 51 mm

   (gerber measured mini mun)

   Total Length_{MAX} = (TL0 + TL1) + TL2 + TL3 + TL4 + TL5_{MAX} = 64 mm

   (gerber measured maxinum)

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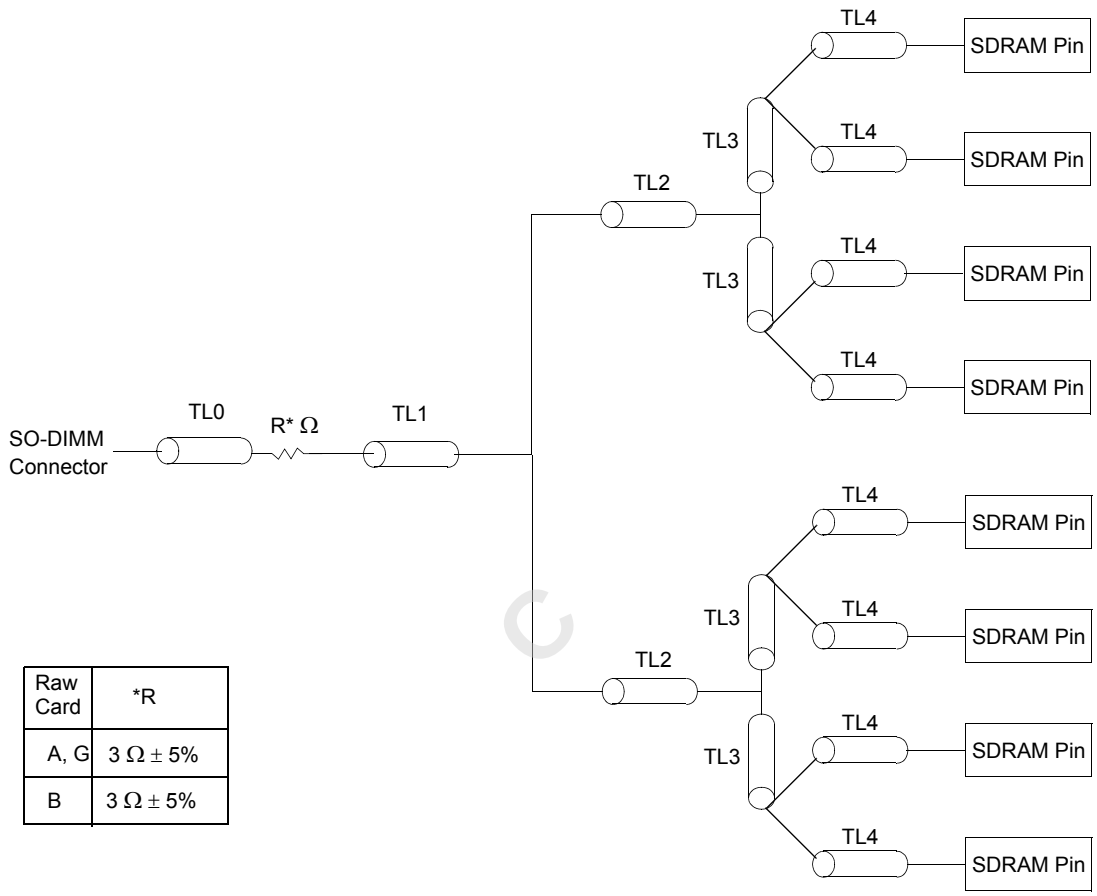
Address/Command Net Structures Ax, BAx, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$ (Raw cards A, B and G)

Net Structure Routing for Address and Command



Raw Card	*R
A, G	3 Ω ± 5%
B	3 Ω ± 5%

Trace Lengths for Address and Command Net Structures

	TL0 Outer		TL1 Outer		TL2 Inner		TL3 Inner		TL4 Outer		Notes
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A, G	3.0	3.6	24.9	25.1	19.3	19.7	7.4	8.0	2.0	6.3	1, 2
B	3.2	4.5	27.9	29.3	19.1	19.3	7.8	7.9	2.0	8.4	1, 3

1. All distances are given in mm and should be kept within a tolerance of ± 0.8 mm
2. For Raw Card A, the total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN}$ = TL0 + TL1 + $TL2_{MIN}$ + $TL3_{MIN}$ + $TL4_{MIN}$ = 57.1 *(gerber measured mininmun)*
   Total $Length_{MAX}$ = TL0 + TL1 + $TL2_{MAX}$ + $TL3_{MAX}$ + $TL4_{MAX}$ = 61.8 *(gerber measured maxinum)*
3. For Raw Card B, the total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN}$ = TL0 + TL1+ $TL2_{MIN}$ + $TL3_{MIN}$ + $TL4_{MIN}$ = 59.1 *(gerber measured mininmun)*
   Total $Length_{MAX}$ = TL0 + TL1+ $TL2_{MAX}$ + $TL3_{MAX}$ + $TL4_{MAX}$ = 68.1*(gerber measured maxinum)*

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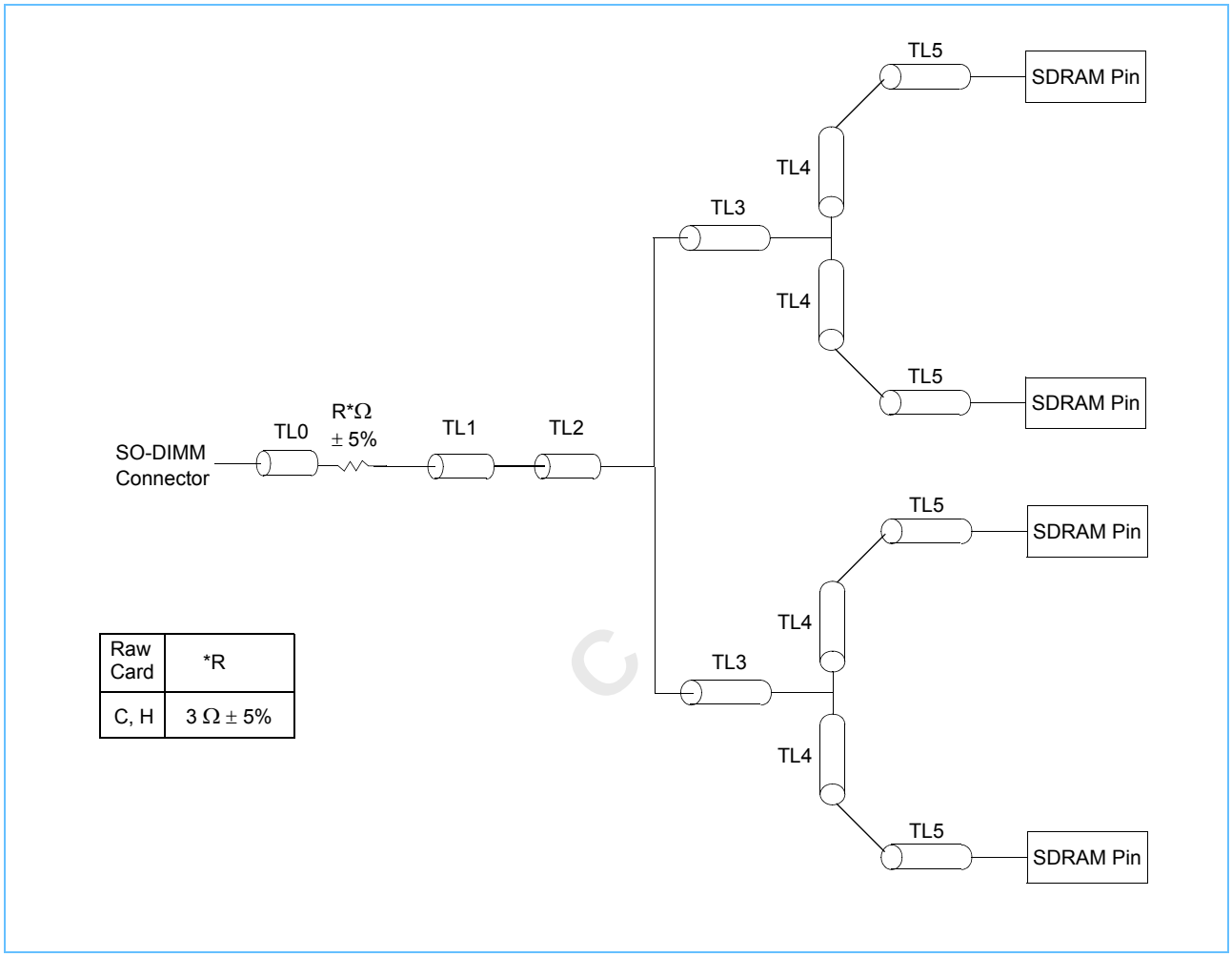
Address/Command Net Structures Ax, BAx, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$ (Raw card C and H)

Net Structure Routing for Address and Command



Trace Lengths for Address and Command Net Structures

	TL0 Outer		TL1 Outer		TL2 Inner		TL1+TL2		TL3 Outer		TL4 Inner		TL5 Outer	
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
C, H	3.0	3.6	1.0	6.0	53.0	58.0	58.9	59.1	18.2	18.5	7.2	7.5	1.0	6.0

1. All distances are given in mm and should be kept within a tolerance of ± 0.8 mm
2. The total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN}$ = TL0+ TL1+ TL2 + $TL3_{MIN}$ + $TL4_{MIN}$ +$TL5_{MIN}$ = 89.0 *(gerber measured mininmun)*
   Total $Length_{MAX}$ = TL0+ TL1+ TL2 + $TL3_{MAX}$ + $TL4_{MAX}$ + $TL5_{MAX}$ = 93.9 *(gerber measured maxinum)*

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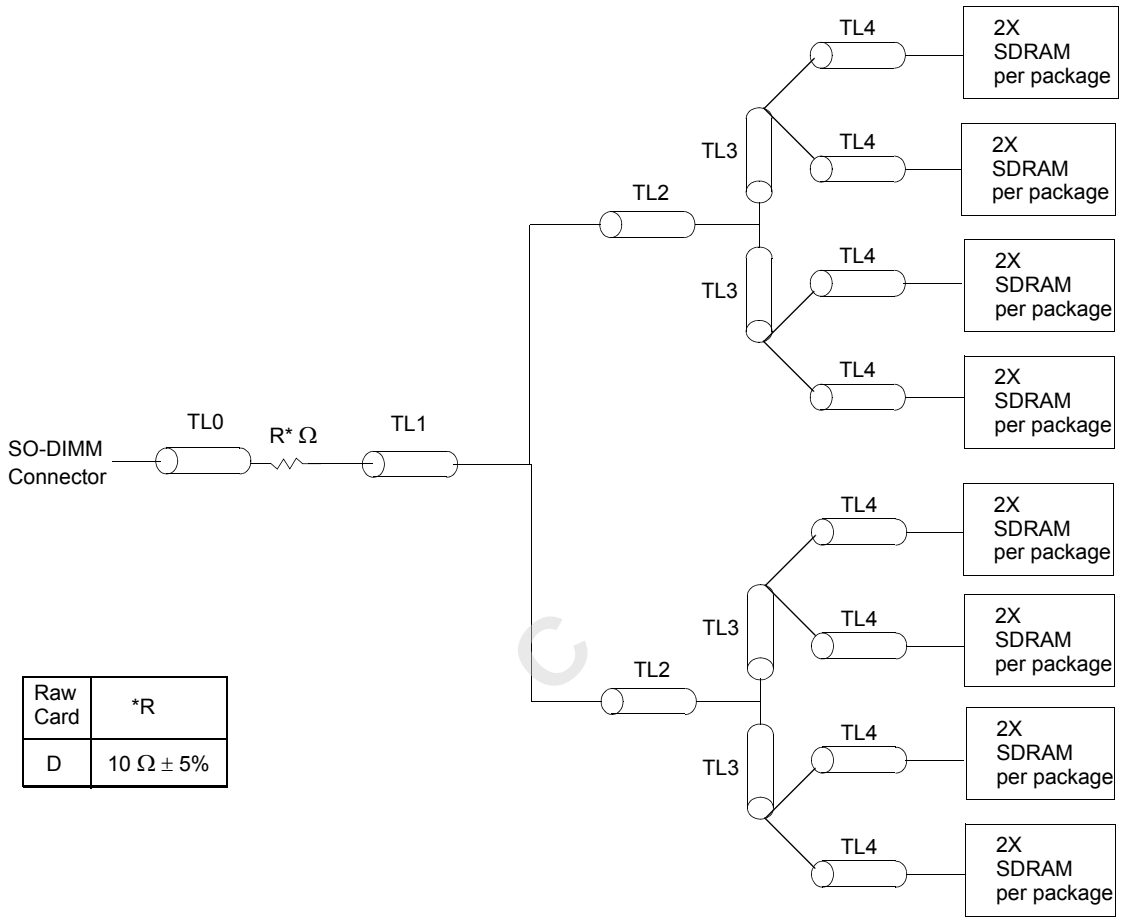
Address/Command Net Structures Ax, BAx, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$ (Raw card D)

Net Structure Routing for Address and Command



Raw Card	*R
D	10 Ω ± 5%

Trace Lengths for Address and Command Net Structures

	TL0 Outer		TL1 Outer		TL2 Inner		TL3 Inner		TL4 Outer	
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
D	3.2	4.5	28.4	29.7	19.2	19.4	7.8	7.9	2.0	10.8

1. All distances are given in mm and should be kept within a tolerance of ± 0.8 mm
2. The total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN}$ = TL0+ TL1+ $TL2_{MIN}$+ $TL3_{MIN}$ + $TL4_{MIN}$ = 61.5 *(gerber measured mininmun)*
   Total $Length_{MAX}$ = TL0+ TL1+ $TL2_{MAX}$+ $TL3_{MAX}$ + $TL4_{MAX}$ = 69.8 *(gerber measured maxinum)*

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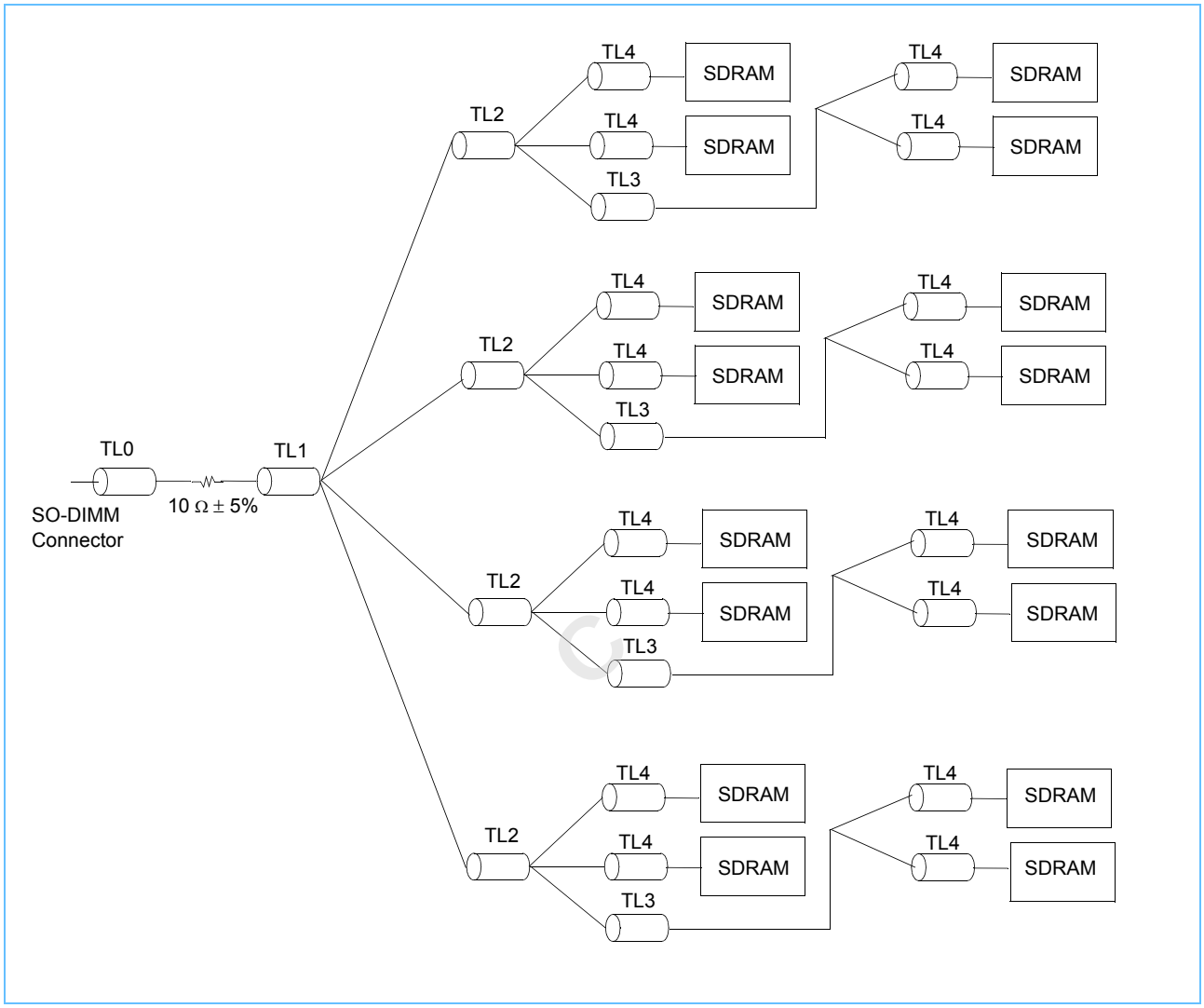
Address/Command Net Structures Ax, BAx, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$ (Raw card E)

Net Structure Routing for Address and Command



Trace Lengths for Address and Command Net Structures

	TL0 Outer		TL1 Inner / Outer		TL2 Inner		TL3 Outer		TL4 Outer		Note
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
E	3.2	4.1	22.6	26.8	13.0	16.0	14.0	14.3	1.7	5.1	1,2,3

1. All distances are given in mm and should be kept within a tolerance of ± 0.8 mm
2. Total of TL0 + TL1 = 30.0 mm
3. The total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN}$ = TL0+ TL1+ $TL2_{MIN}$ + $TL3_{MIN}$+ $TL4_{MIN}$ = 41.3*(gerber measured mininmun)*
   Total $Length_{MAX}$ = TL0+ TL1+ $TL2_{MAX}$ +$TL3_{MAX}$ + $TL4_{MAX}$ = 61.2*(gerber measured maxinum)*

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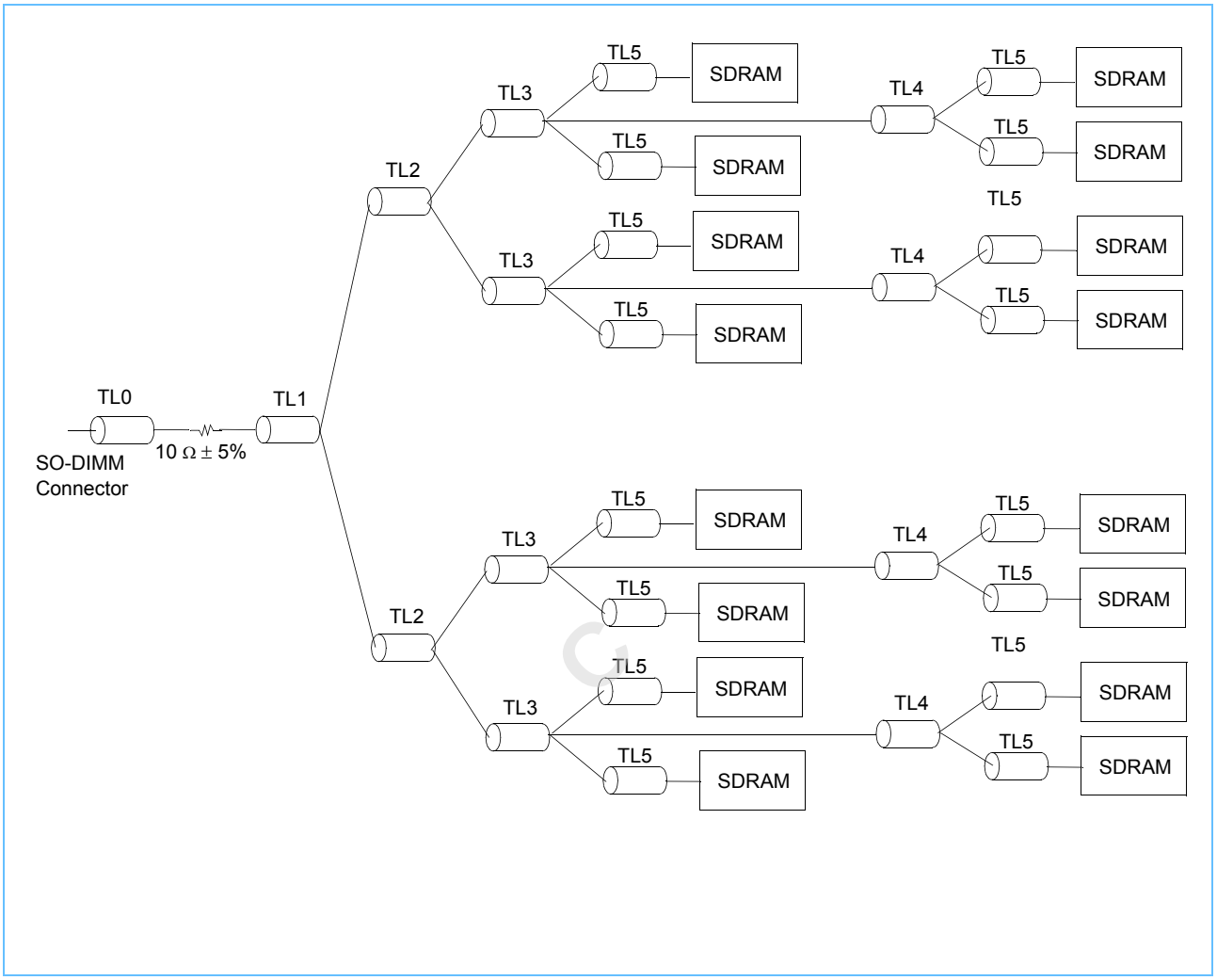
Address/Command Net Structures Ax, BAx, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$ (Raw card F)

Net Structure Routing for Address and Command



Trace Lengths for Address and Command Net Structures

	TL0 Outer		TL1 Inner / Outer		TL2 Inner		TL3 Inner		TL4 Inner		TL5 Outer		Note
Raw Card	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
F	3.2	5.9	24.1	26.8	19.5	19.5	11.3	11.3	13.0	13.3	2.0	7.2	1, 2, 3, 4

1. All distances are given in mm and should be kept within a tolerance of ± 0.1 mm
2. Total of TL0 + TL1 = 30.0 mm
3. TL0 and TL1 are given for lines without any via. For example, if there is a via on the the transmission line, then the line length has to be reduced by 2.54 mm, and if there are two vias on the TL, then the line length has to be reduced by 5.08 mm.
4. The total routed trace lengths from card edge to DRAM pin are defined by the following equations:
   Total $Length_{MIN}$ = (TL0+ TL1) + TL2 + TL3 + $TL5_{MIN}$ = 62.8 mm *(gerber measured mininmun)*
   Total $Length_{MAX}$ = (TL0+ TL1) + TL2 + TL3 + TL4 + $TL5_{MAX}$ = 81.5mm *(gerber measured maxinum)*

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Cross Section Recommendations

The DDR2 SO-DIMM printed circuit board design uses six-layers of glass epoxy material. Layers 2 and 4 are divided so adjacent signal layers maintain a constant Vss or Vdd reference. All data is referenced to Vss and all address/command are referenced to Vdd. The PCB stackup must be designed with 4 mil wide traces. The required board impedance is 60 Ω ± 10%.

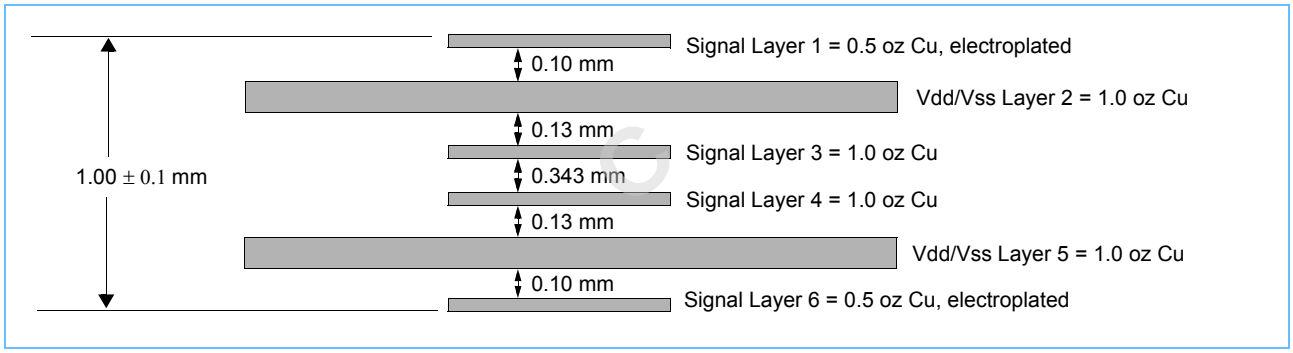
Note: The PCB edge connector contacts shall be gold-plated; chamfered edges are optional.

PCB Electrical Specifications

Parameter	Min	Max	Units
Trace velocity: S0 (outer layers)	5.5	6.7	ps/mm
Trace velocity: S0 (inner layers)	6.5	7.6	ps/mm
Trace impedance: Z0 (all layers)	54	66	Ohms

Example Layer Stackup

Raw Card A, B, C, D, G and H



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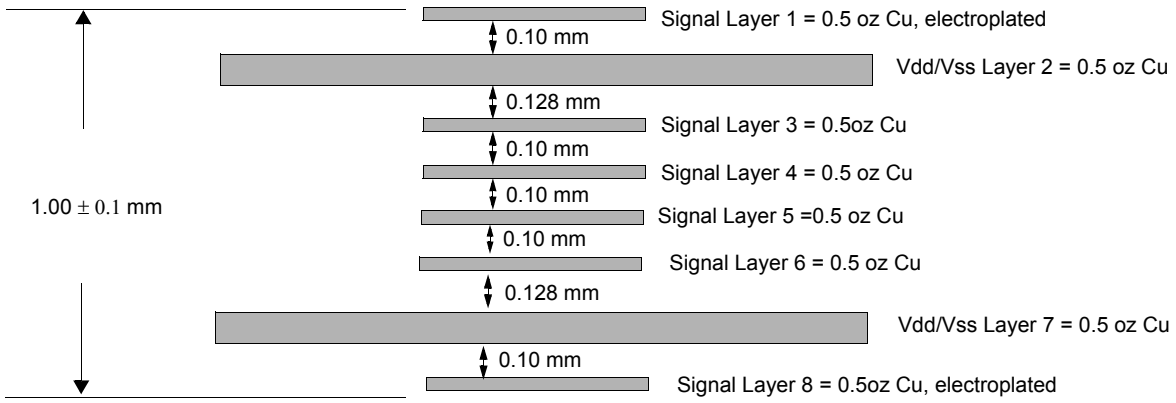
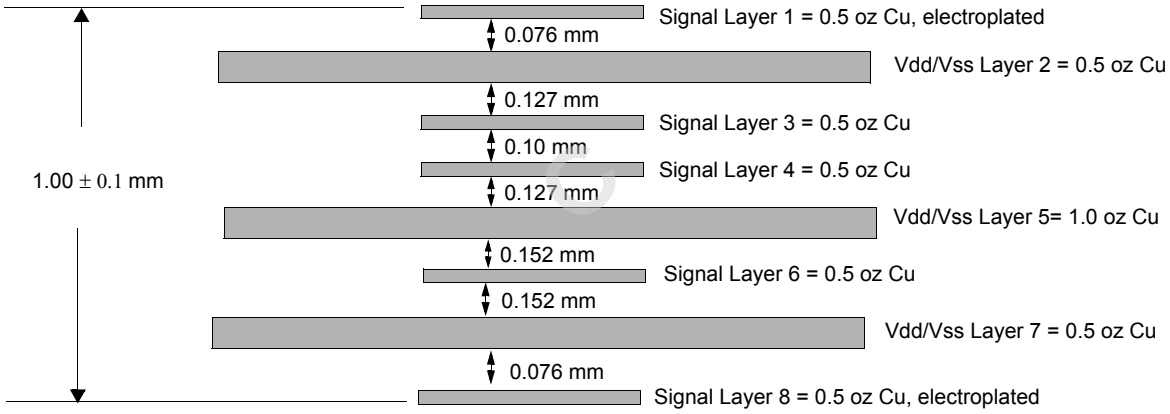
Raw Card E



Raw Card F



Component Types and Placement

Components shall be surface mounted on both sides of the PCB and positioned on the PCB to meet the minimum and maximum trace lengths required for DDR2 SDRAM signals. Bypass capacitors for DDR2 SDRAM devices, especially for VREF, must be practically located near the device power pins.

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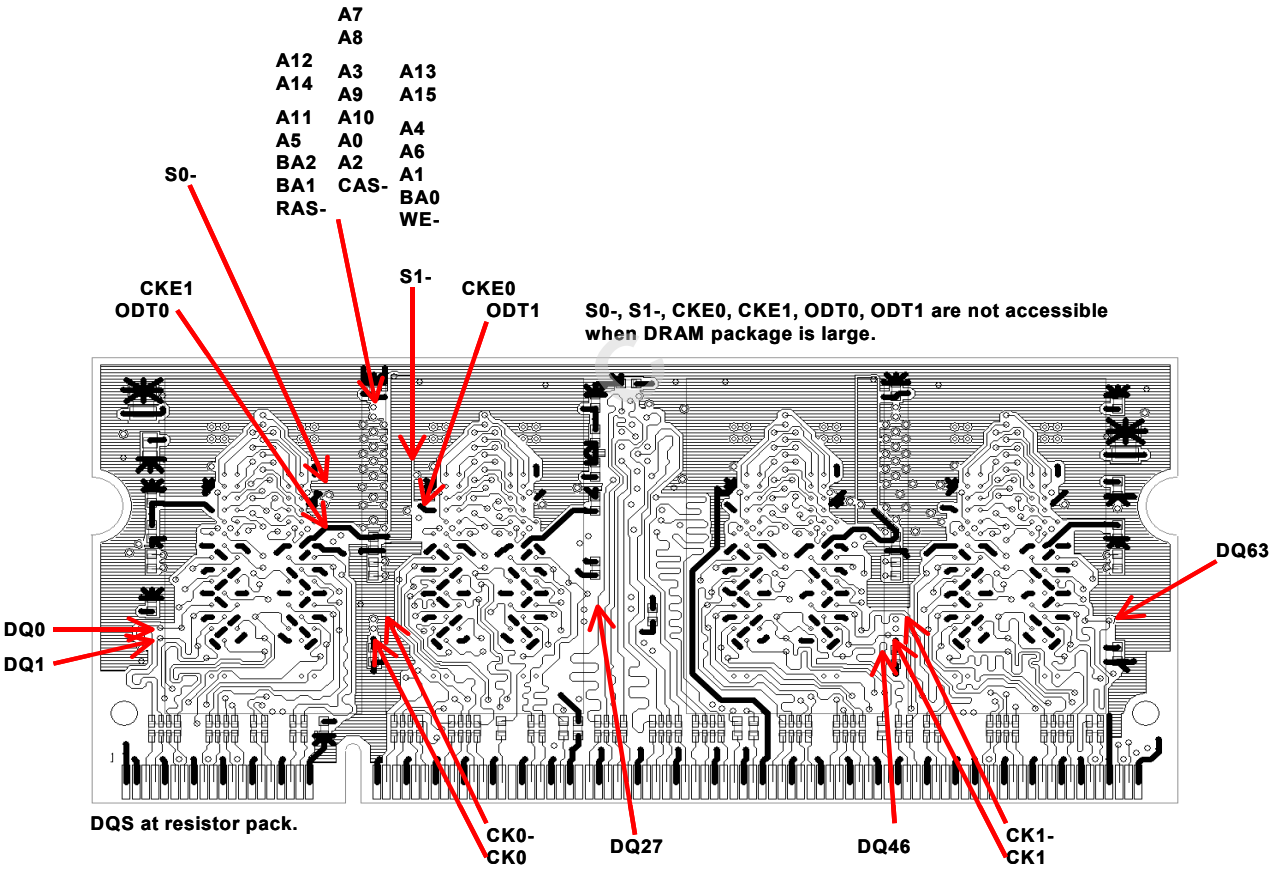
7. Test Points

All DDR2 components are in BGA packages which makes the package pads inaccessible for probing during system development. The DDR2 SO-DIMMs have test points identified to make initial evaluation easier. In some cases test pads have been added and in other cases existing vias are used as test points. An effort has been made to provide testability on some signals in all signal groups but 100% coverage is not possible.

Raw Card A Test Points

The test point coverage for RC A is on both the front and back sides of the card. The following figures detail the locations of the test points.

Raw Card A Front View



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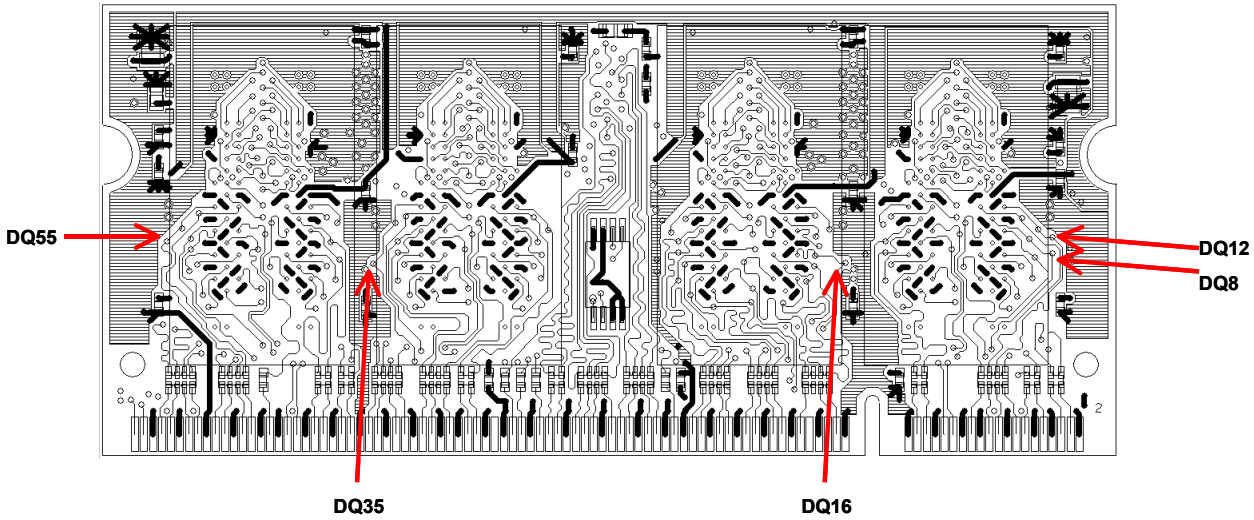
Raw Card A Back View



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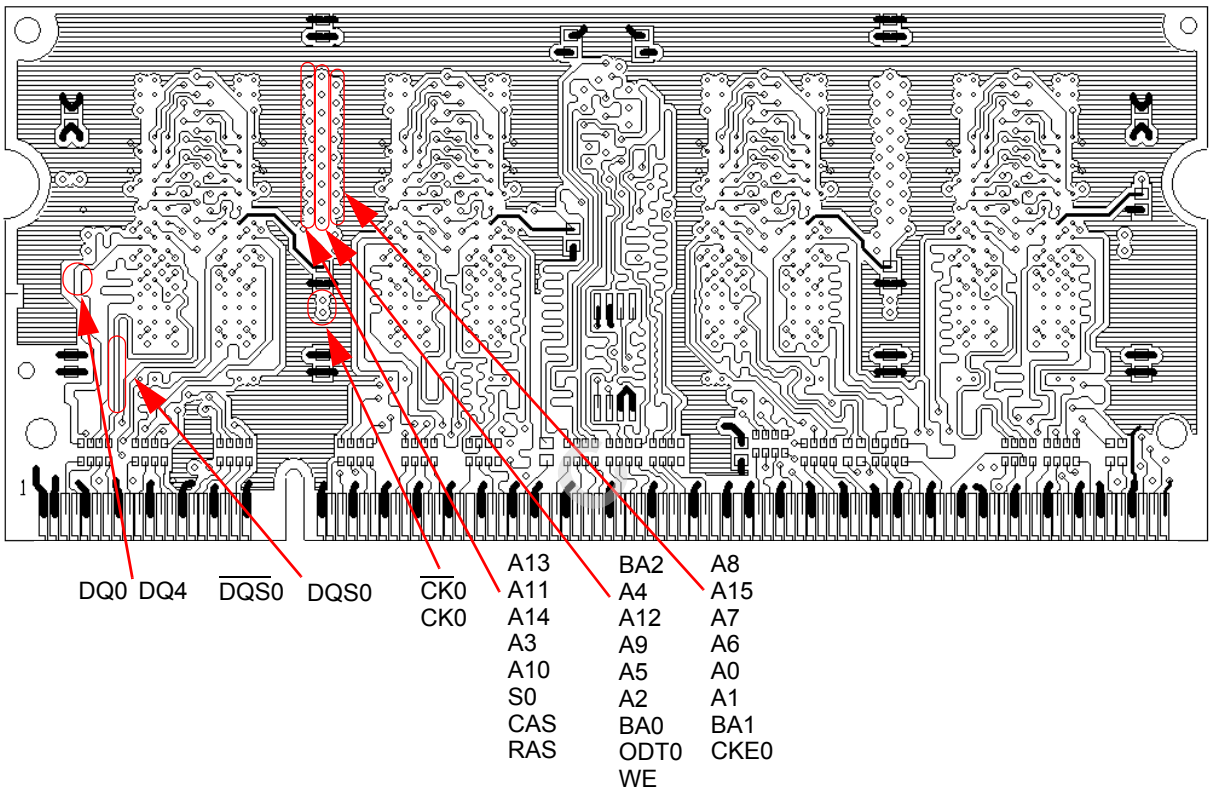
Raw Card B Test Points

The test point coverage for RC B is very complete for the address, command and control nets. These signals are easily probed at the vias near the edge of the component. Test coverage for the data bus is not as complete. Two data signals and the associated data strobe signals have been make accessible by opening a hole in solder mask for the probe points.

Raw Card B Front View



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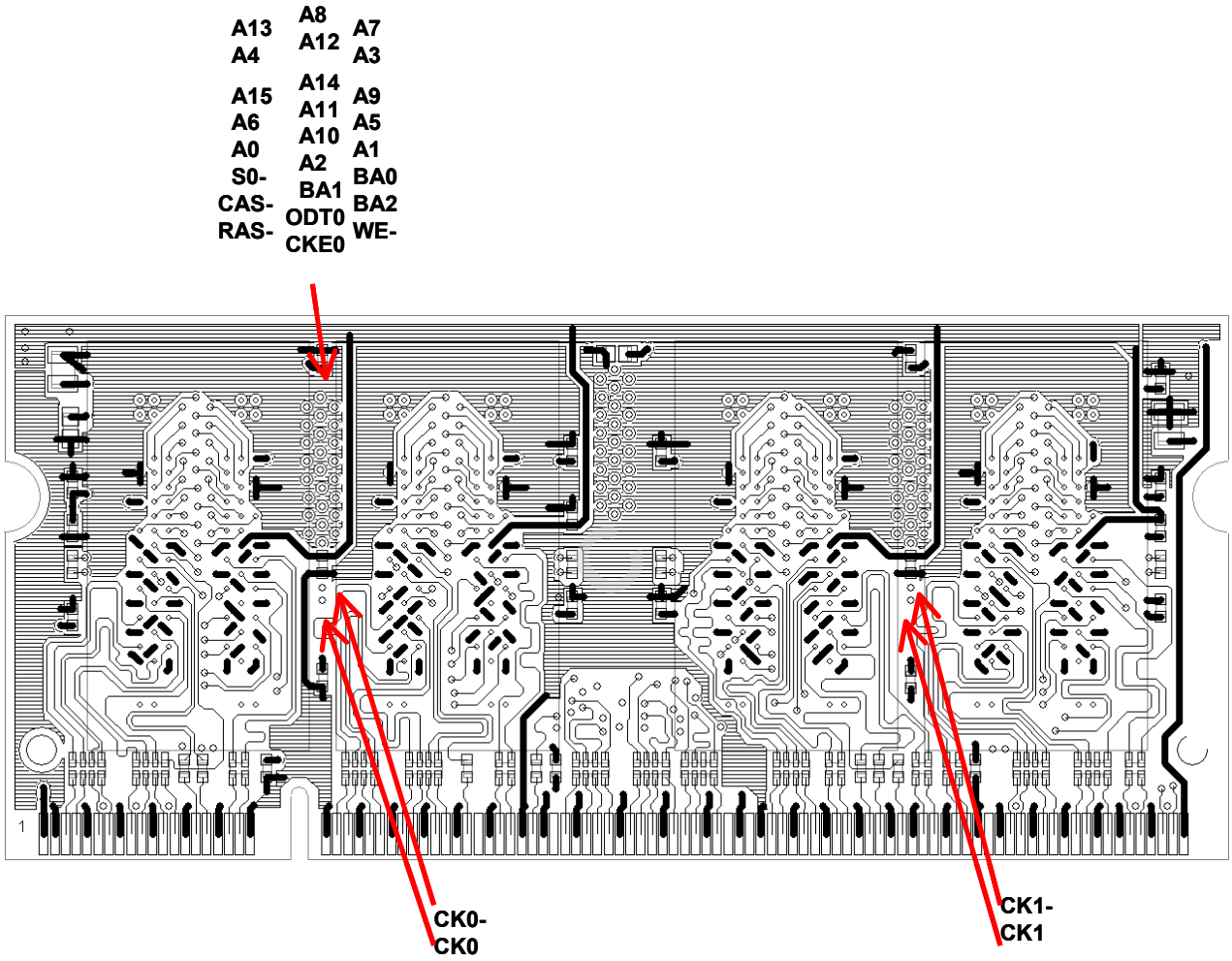
Raw Card C Test Points

The test point coverage for RC C is on both the front and back sides of the card. The following figures detail the locations of the test points.

Raw Card C Front View



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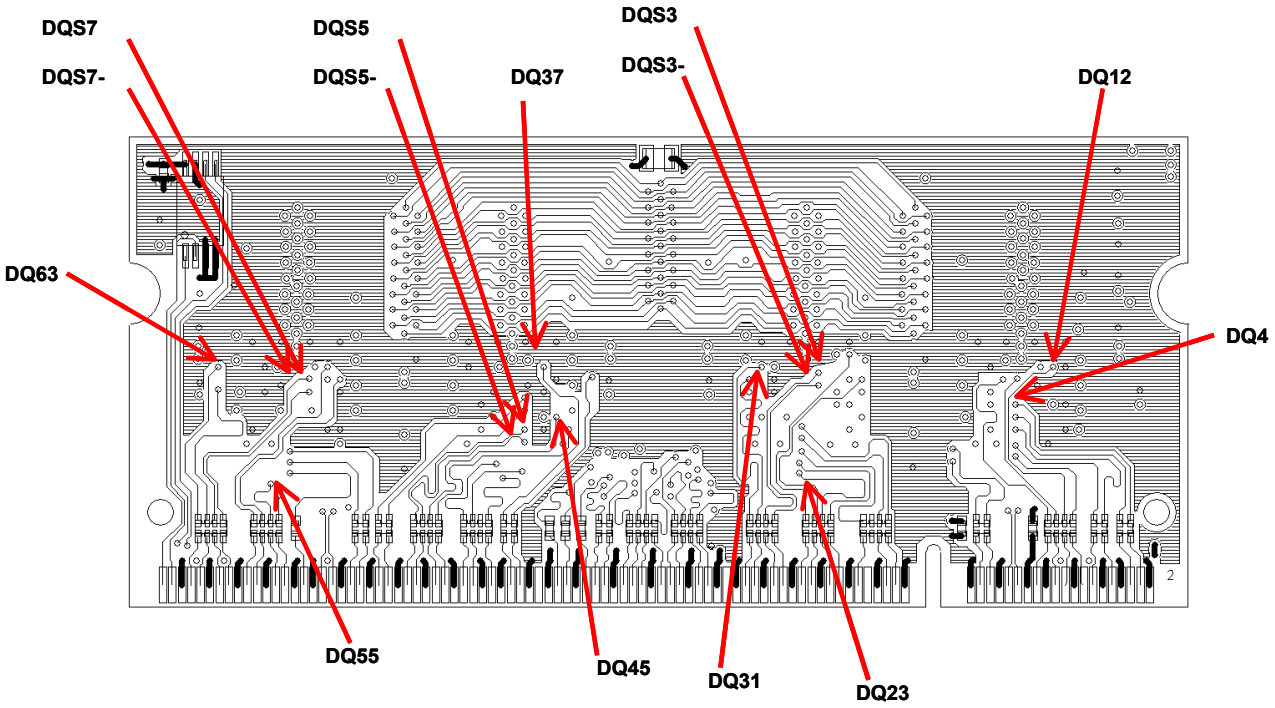
Raw Card C Back View



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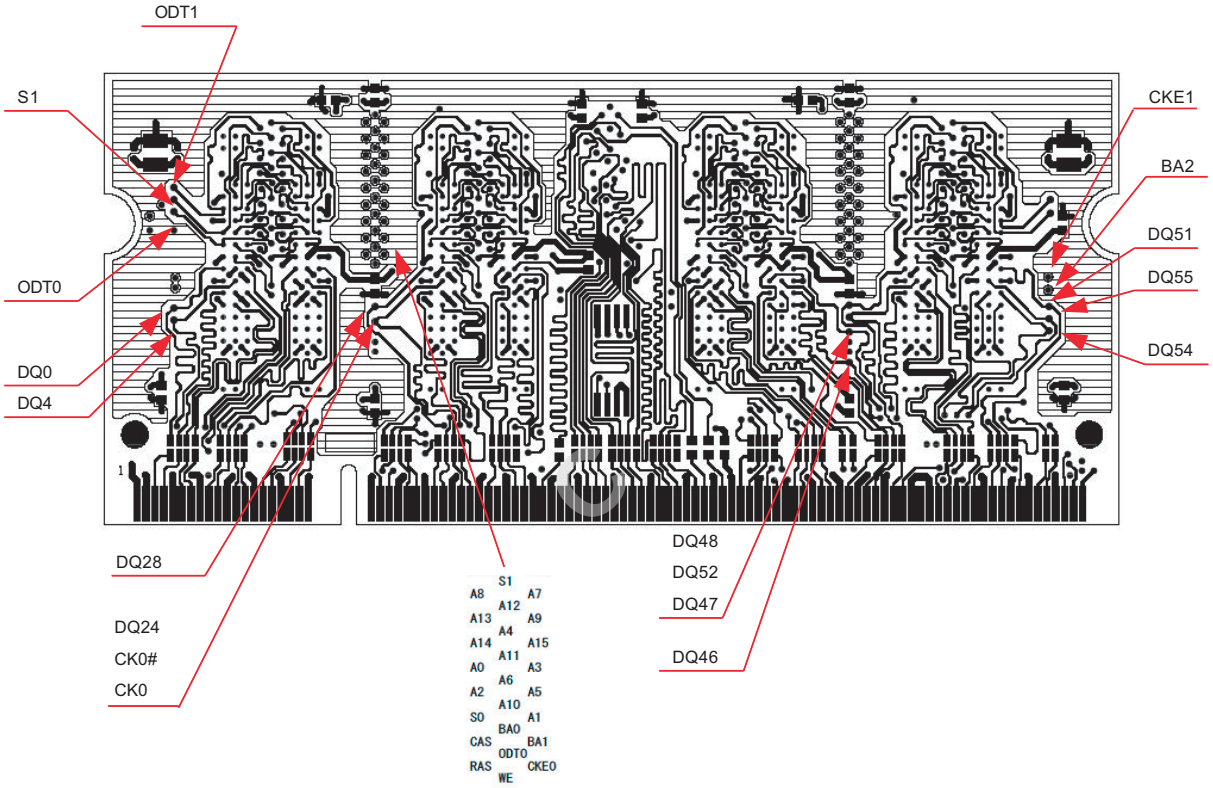
Raw Card D Test Points

The test point coverage for RC D is very complete for the address, command and control nets. These signals are easily probed at the vias near the edge of the component. Test coverage for the data bus is not as complete. Two data signals and the associated data strobe signals have been make accessible by opening a hole in solder mask for the probe points.

Raw Card D Front View



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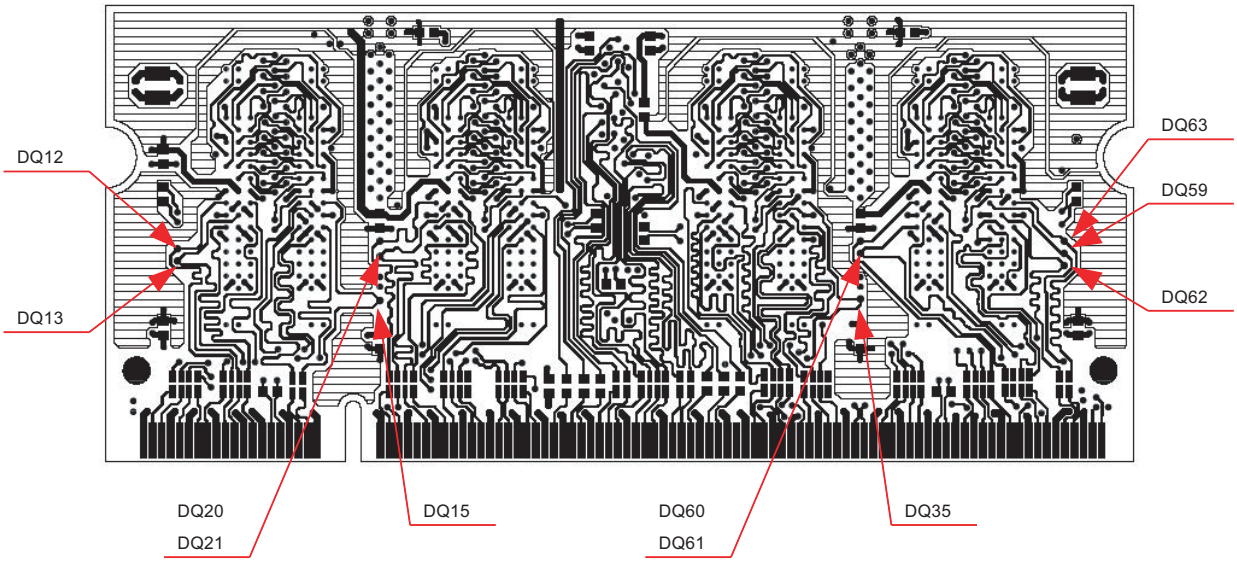
Raw Card D Back View



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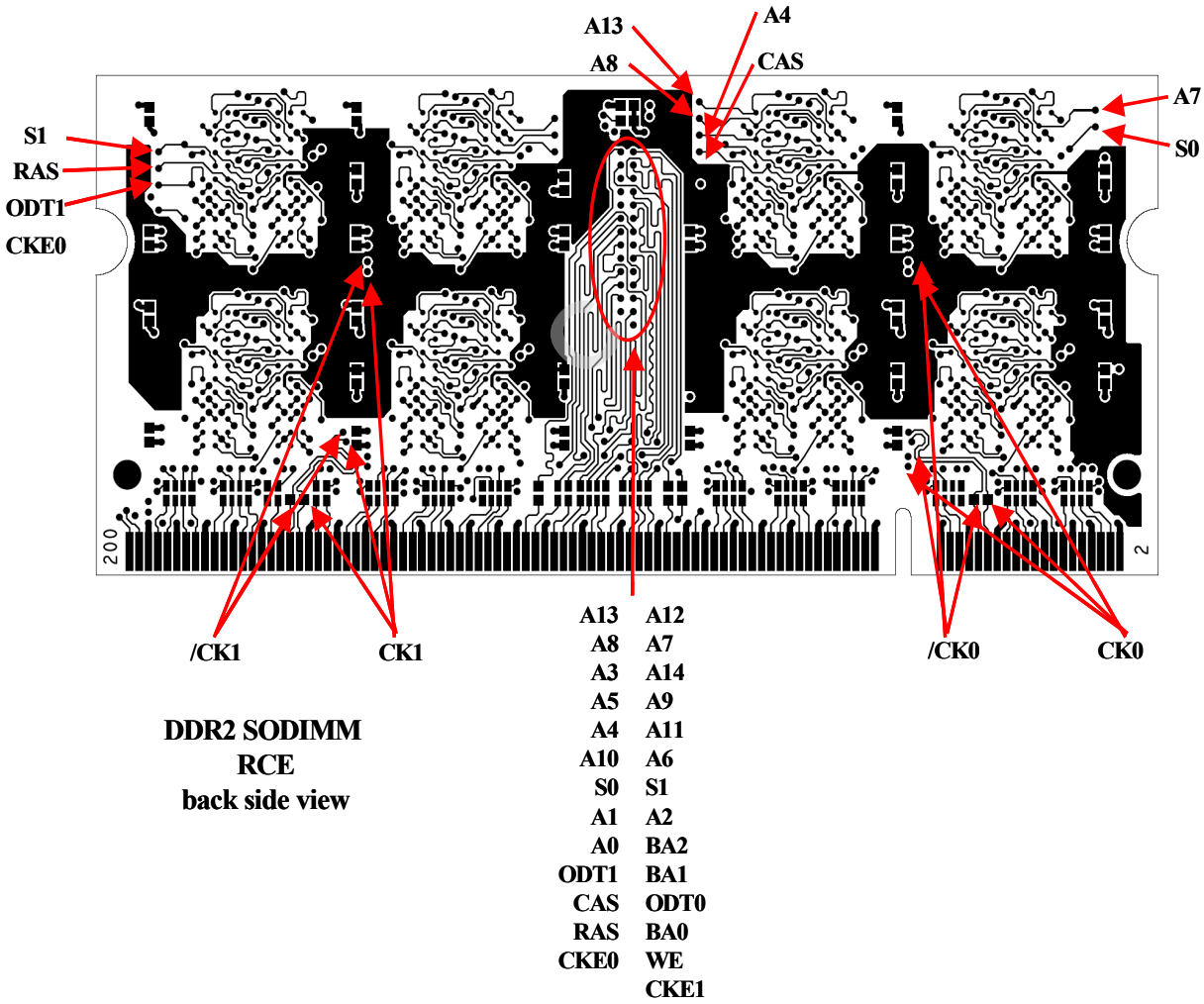
Raw Card E Test Points

The test point coverage for RC E is very complete for the address, command and control nets. These signals are easily probed at the vias near the edge of the component. Test coverage for the data bus is not as complete. Two data signals and the associated data strobe signals have been make accessible by opening a hole in solder mask for the probe points.

Raw Card E Back View



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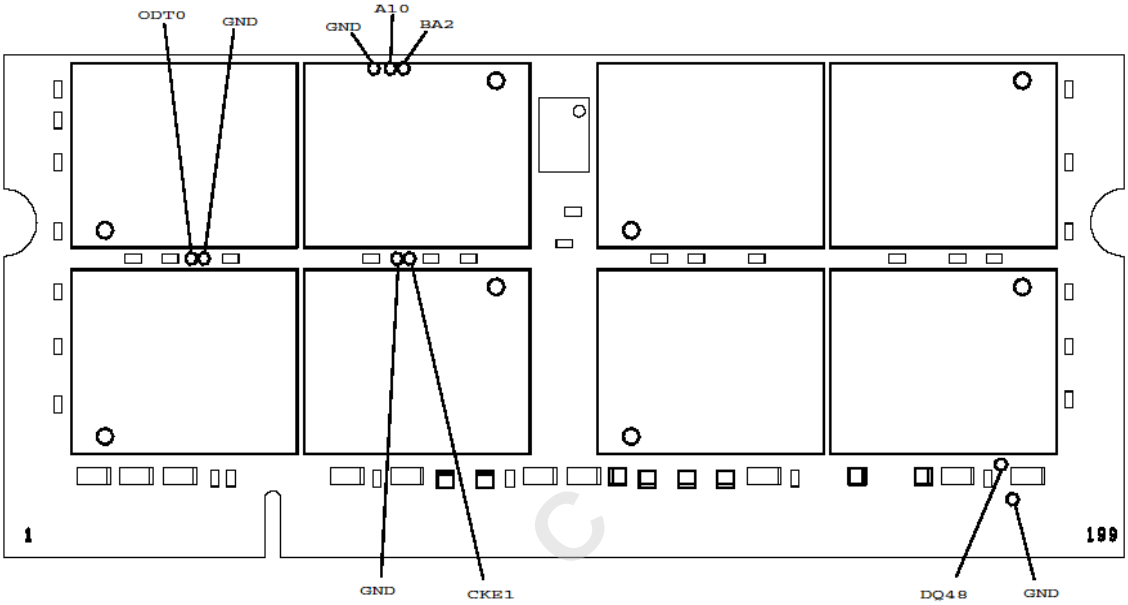
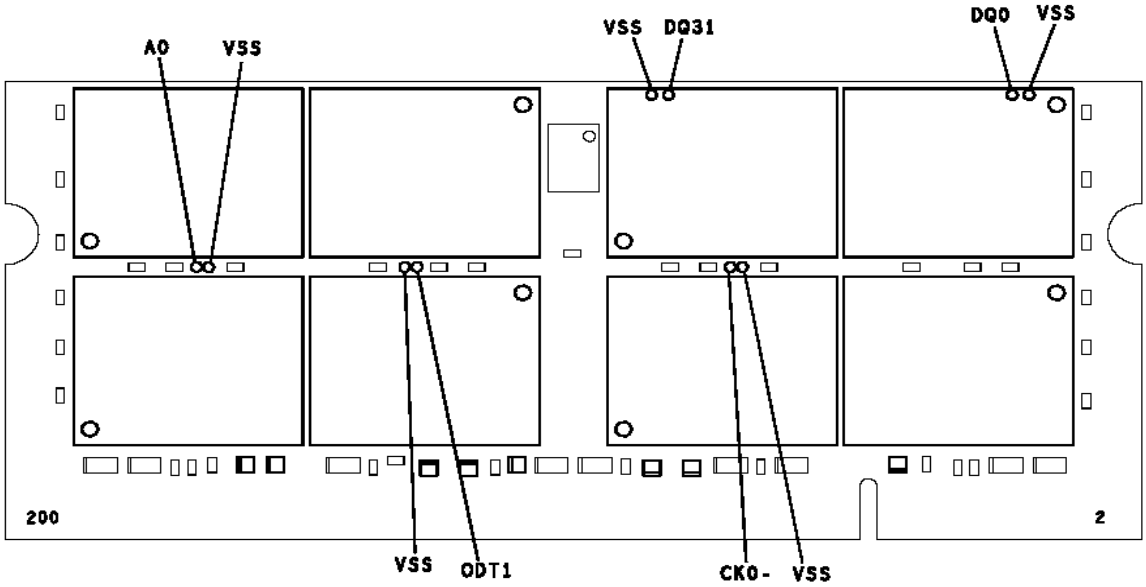
Raw Card F Test Points

The test point coverage for RC F is very complete for the address, command and control nets. These signals are easily probed at the test points near the edge of the component.

Raw Card F Front View



Raw Card F Back View



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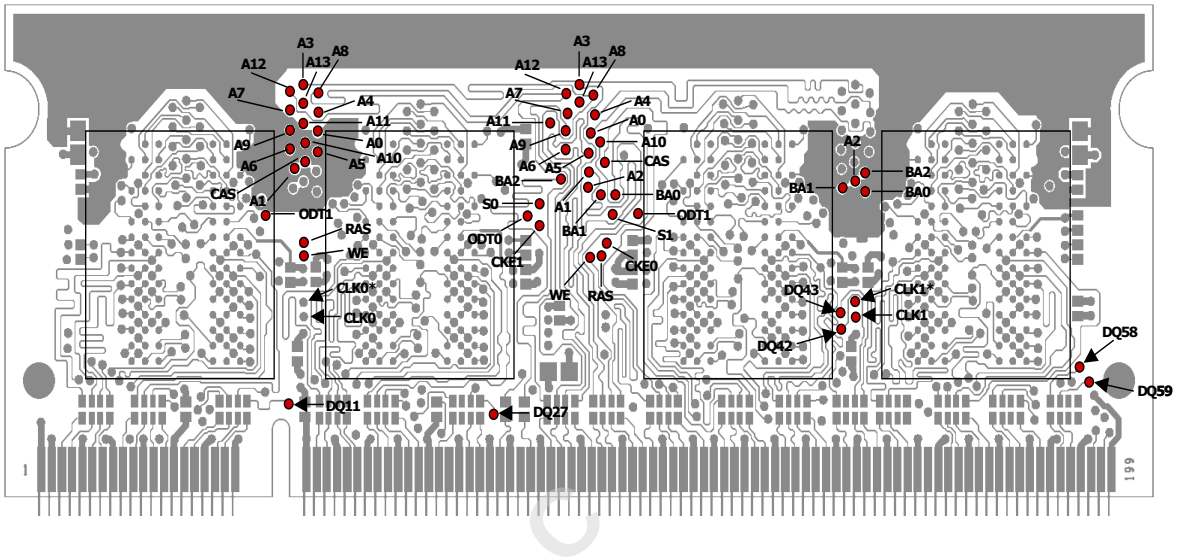
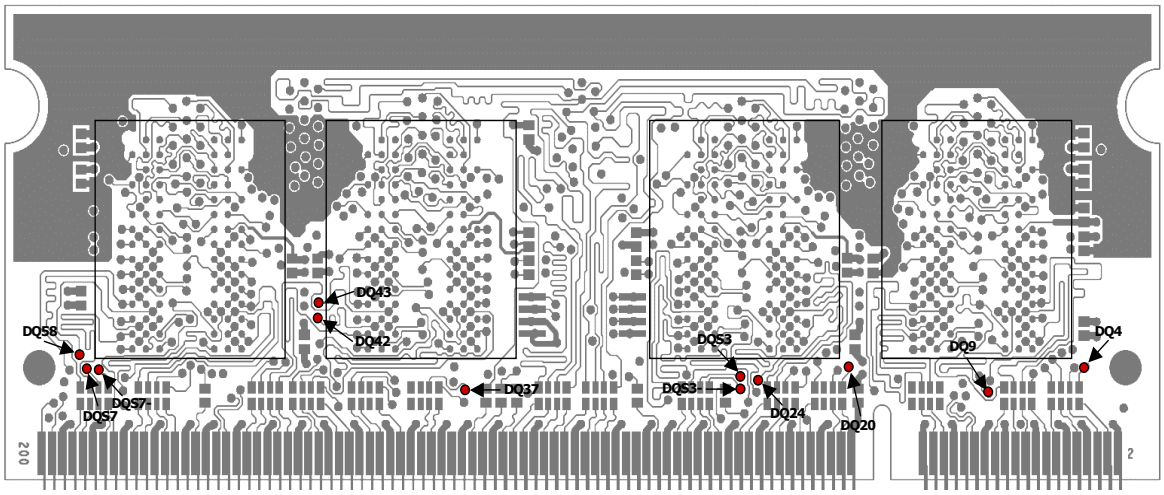
Raw Card G Test Points

Raw Card G Front View



Raw Card G Back View



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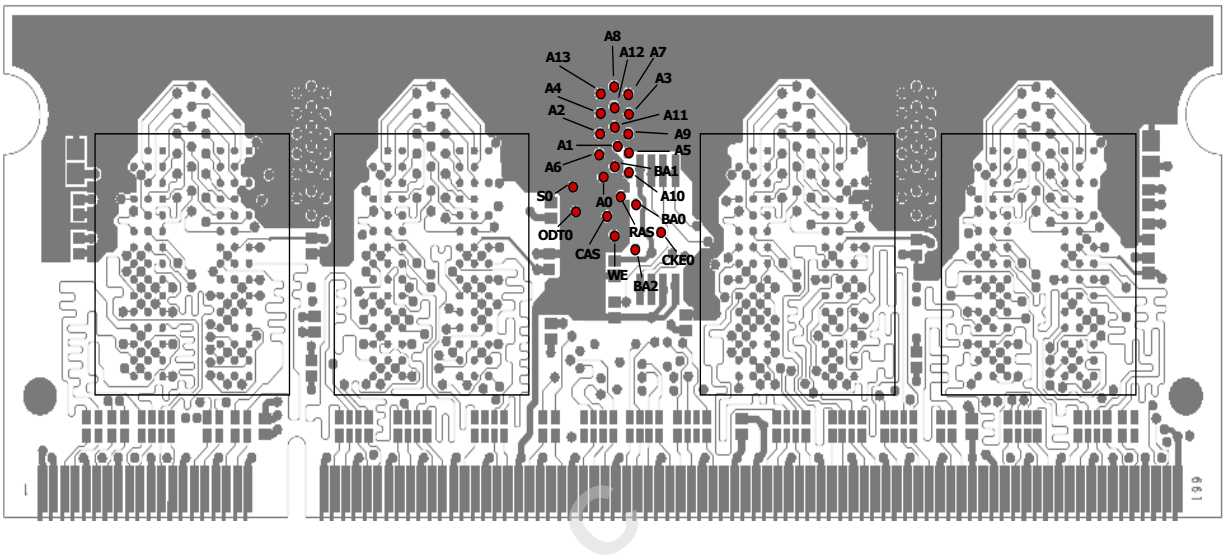
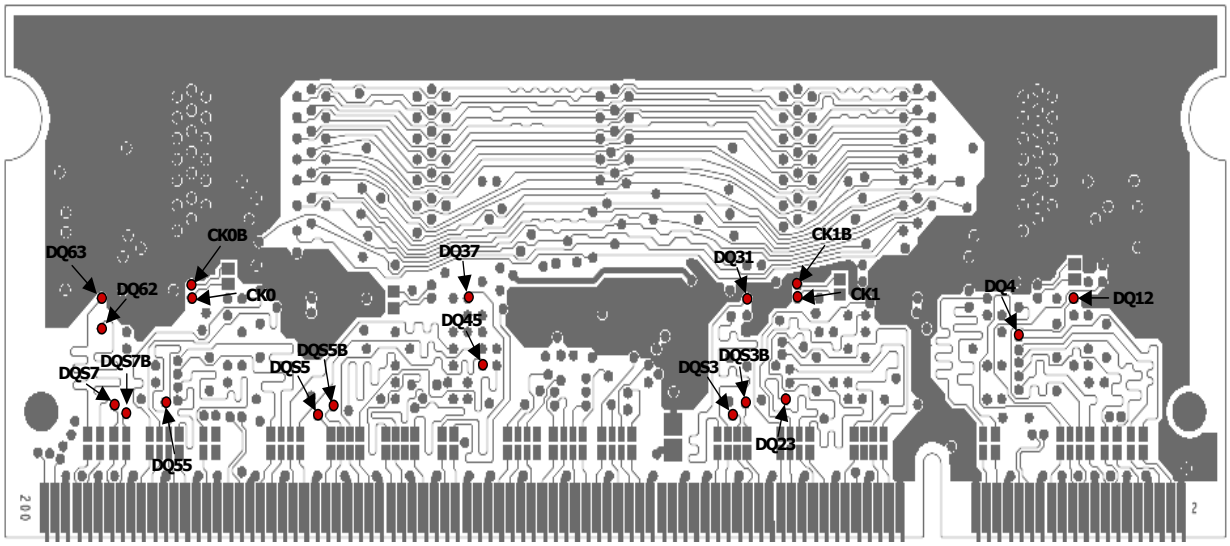
Raw Card H Test Points

Raw Card H Front View



Raw Card H Back View



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8. Serial Presence Detect Definition

The Serial Presence Detect (SPD) function MUST be implemented on the DDR2 SDRAM Unbuffered SO-DIMM. The component used and the data contents must adhere to the most recent version of the JEDEC DDR2 SDRAM SPD Specifications. Please refer to this document for all technical specifications and requirements of the serial presence detect devices.

The following table is intended to be an **example** of a typical PC2-5300 SO-DIMM. SPD values indicating different SO-DIMM performance characteristics will be utilized based on specific characteristics of the SDRAMs or SO-DIMMs. This example assumes:

- Module Organization: 256 MB
- Device Composition: 32Mx8
- Device Package: FBGA
- Module Physical Ranks: 2
- CAS latency: 4(DDR2-667)

Serial Presence Detect Data Example (Part 1 of 3)

Byte # (dec)	Byte # (hex)	Description	SPD Entry Value	Serial PD Data Entry (Hexadecimal)	Notes
			DDR2-667	DDR2-667	
0	00	Number of Serial PD Bytes written during production	128	80	1
1	01	Total Number of Bytes in Serial PD device	256	08	2
2	02	Fundamental Memory Type (FPM, EDO, SDRAM, DDR, DDR2, ...)	DDR2 SDRAM	08	
3	03	Number of Row Addresses on Assembly	13	0D	
4	04	Number of Column Addresses on Assembly	10	0A	
5	05	Number of DIMM Ranks	Module height: 30 mm, Planar, card on card: no, 1 Rank	60	
6	06	Data Width of this Assembly	x64	40	
7	07	Reserved	Undefined	00	
8	08	Voltage Interface Level of this assembly	SSTL 1.8V	05	
9	09	SDRAM Cycle Time at maximum supported $\overline{CAS}$ latency (CL), CL = X	3.75 ns	30	3
10	0A	SDRAM Access from Clock	+/-0.45 ns	45	
11	0B	DIMM configuration type (Non-parity, or ECC)	Non-Parity	00	
12	0C	Refresh Rate/Type	7.8 us	82	3,4
13	0D	Primary SDRAM Width	x8	08	
14	0E	Error Checking SDRAM Width	NA	00	
15	0F	Reserved	Undefined	00	
16	10	SDRAM device attributes: Burst lengths supported	4,8	0C	

1. This will typically be programmed as 128 bytes.
2. This will typically be programmed as 256 bytes.
3. From Data sheet.
4. High order bit is self refresh "flag". If set to "1", the assembly supports self refresh.
5. These are optional, in accordance with the JEDEC specification.

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Serial Presence Detect Data Example (Part 2 of 3)

Byte # (dec)	Byte # (hex)	Description	SPD Entry Value	Serial PD Data Entry (Hexadecimal)	Notes
			DDR2-667	DDR2-667	
17	11	SDRAM device attributes: Number of Banks on SDRAM device	4	04	3
18	12	SDRAM device attributes: $\overline{CAS}$ Latency	5,4,3	38	3
19	13	Reserved	Undefined	00	0
20	14	DIMM type information	SO-DIMM	04	
21	15	SDRAM Module Attributes	Normal DIMM	00	
22	16	SDRAM device attributes: General	Suports weak drivet	01	
23	17	Minimum Clock Cycle at CL=X -1	3.75 ns	3D	3
24	18	Maximum Data Access Time (t_{AC}) from Clock at CL=X -1	500 ps	50	3
25	19	Minimum Clock Cycle Time at CL=X-2	5 ns	50	3
26	1A	Maximum Data Access Time (t_{AC}) from Clock at CL=X-2	600 ps	60	3
27	1B	Minimum Row Precharge Time (t_{RP})	15.0 ns	3C	3
28	1C	Minimum Row Active to Row Active delay (t_{RRD})	7.5 ns	1E	3
29	1D	Minimum $\overline{RAS}$ to $\overline{CAS}$ delay (t_{RCD})	15.0 ns	2D	3
30	1E	Minimum Active to Precharge Time (t_{RAS})	45.0 ns	2D	3
31	1F	Module Rank Density	256 MB	40	
32	20	Address and Command input Setup Time Before Clock (t_{IS})	250 ps	25	3
33	21	Address and Command input Hold Time After Clock (t_{IH})	500 ps	27	3
34	22	Data Input Setup Time Before Strobe (t_{DS})	300 ps	05	3
35	23	Data Input Hold Time After Strobe (t_{DH})	300 ps	17	3
36	24	Write recovery time (t_{WR})	15.0 ns	3C	3
37	25	Internal write to read command delay (t_{WTR})	7.5 ns	1E	3
38	26	Internal read to precharge command delay (t_{RTP})	7.5 ns	1E	3
39	27	Memory analysis probe characteristics	Undefined	00	
40	28	Extension of byte 41 and 42.	41, 42 extension	00	
41	29	SDRAM device minimum active to active/auto refresh time (t_{RC})	60 ns	3C	3
42	2A	SDRAM device minimum auto-refresh to active/auto refresh command period (t_{RFC})	75 ns	4B	3
43	2B	SDRAM device maximum device cycle time (t_{CKmax})	8.0 ns	80	3

1. This will typically be programmed as 128 bytes.
2. This will typically be programmed as 256 bytes.
3. From Data sheet.
4. High order bit is self refresh "flag". If set to "1", the assembly supports self refresh.
5. These are optional, in accordance with the JEDEC specification.

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Serial Presence Detect Data Example (Part 3 of 3)

Byte # (dec)	Byte # (hex)	Description	SPD Entry Value	Serial PD Data Entry (Hexadecimal)	Notes
			DDR2-667	DDR2-667	
44	2C	SDRAM device maximum skew between DQS and DQ signals	300 ps	19	3
45	2D	SDRAM device maximum read data hold skew factor (t_{QHS})	400 ps	23	3
46	2E	PLL relock time	Undefined	00	3
47 - XX	2F-XX	Reserved for IDD in SPD	Undefined	00	
XX - 61	XX-3D	Reserved	Undefined	00	
62	3E	SPD revision	1.1	1.1	
63	3F	Checksum for bytes 0-62	Checksum Data	cc	
64-71	40-47	Manufacturers' JEDEC ID Code			
72	48	Module Manufacturing Location			
73-90	49-5A	Module Part Number			5
91-92	5B-5C	Module Revision Code			5
93-94	5D-5E	Module Manufacturing Date	Year/Week code	yyww	5
95-98	5F-62	Module Serial Number	Serial Number	ssssssss	
99-127	63-7F	Manufacturer's Specific Data	Undefined	XX	5
128-255	80-FF	Open for Customer Use	Undefined	XX	

1. This will typically be programmed as 128 bytes.
2. This will typically be programmed as 256 bytes.
3. From Data sheet.
4. High order bit is self refresh "flag". If set to "1", the assembly supports self refresh.
5. These are optional, in accordance with the JEDEC specification.

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9. Product Label

The following label shall be applied to all DDR2 memory modules to fully describe the key attributes attributes of the module. The label can be in the form of a stick-on label, silk screened onto the assembly, or using an marked using an alternate customer-readable format. A minimum font size of 8 points should be printed in used, and the number can be printed in one or more rows on the label. Hyphens may be dropped when lines are split, or when font changes sufficiently separate fields.

ggggg eRxff PC2-wwwwm-abc-dd-ef

Where:

ggggg = Module total capacity, in bytes
256MB, 512MB, 1GB, 2GB, 4GB, etc.
eR = Number of ranks of memory installed
1R = 1 rank of DDR2 SDRAM installed
2R = 2 ranks
xff = Device organization (bit width) of DDR2 SDRAMs used on this assembly
x4 = x4 organization (4 DQ lines per SDRAM)
x8 = x8 organization
x16 = x16 organization
wwww = Module bandwidth in MB/s
3200 = 3.20 GB/s
4200 = 4.26 GB/s
5300 = 5.33 GB/s
6400 = 6.40 GB/s
m = Module Type
E = Unbuffered DIMM ("UDIMM"), with ECC (x72 bit module data bus)
F = Fully Buffered DIMM ("FB-DIMM")
M = Micro-DIMM
N = Mini-Registered DIMM ("Mini-RDIMM"), no address/command parity function
P = Registered DIMM ("RDIMM"), with address/command parity function
R = RDIMM, no address/command parity function
S = Small Outline DIMM ("SO-DIMM")
U = Unbuffered DIMM ("UDIMM"), no ECC (x64 bit module data bus)
a = DDR2 SDRAM CAS Latency, in clocks at maximum operating frequency
b = DDR2 SDRAM minimum tRCD specification, in clocks at maximum operating frequency
c = DDR2 SDRAM minimum tRP specification, in clocks at maximum operating frequency
dd = JEDEC SPD Revision Encoding and Additions level used on this DIMM
e = Reference design file used for this design (if applicable)
A = Reference design for raw card ’A’ is used for this assembly
B = Reference design for raw card ’B’ is used for this assembly
Z = None of the reference designs were used for this assembly
f = Revision number of the reference design used
0 = Initial release
1 = First revision
2 = Second revision
P = Pre-release or Engineering sample
Z = To be used when field e = Z

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Examples, single font:

1GB 2Rx4 PC2-3200R-333-11-D2
is a 1GB DDR2 Registered DIMM, without address/command parity function, using 2 ranks of x4 SDRAMs operational to PC2-3200 performance with CAS Latency = 3, t_{RCD} = 3, t_{RP} = 3, using JEDEC SPD revision 1.1, raw card reference design file D revision 2 used for the assembly

1GB 2Rx4 PC2-3200P-333-11-D2
is a 1GB DDR2 Registered DIMM, with address/command parity function, using 2 ranks of x4 SDRAMs operational to PC2-3200 performance with CAS Latency = 3, t_{RCD} = 3, t_{RP} = 3, using JEDEC SPD revision 1.1, raw card reference design file D revision 2 used for the assembly

1GB 2Rx4 PC2-3200P-333-11-DP
is a 1GB DDR2 Registered DIMM, with address/command parity function, using 2 ranks of x4 SDRAMs operational to PC2-3200 performance with CAS Latency = 3, t_{RCD} = 3, t_{RP} = 3, using JEDEC SPD revision 1.1, raw card reference design file D pre-release engineering sample layout used for the assembly

1GB 1Rx4 PC2-3200F-333-10-C0
is a 1GB DDR2 Fully Buffered DIMM, using 1 rank of x4 SDRAMs operational to PC2-3200 performance with CAS Latency = 3, t_{RCD} = 3, t_{RP} = 3, using JEDEC SPD revision 1.0 (encoding type FB-DIMM), raw card reference design file C initial release used for the assembly

256MB 1Rx16 PC2-4200M-544-11-B1
is a 256MB DDR2 Micro-DIMM using 1 rank of x16 SDRAMs operational to PC2-4200 performance with CAS Latency = 5, t_{RCD} = 4, t_{RP} = 4, using JEDEC SPD revision 1.1, raw card reference design file B revision 1 used for the assembly

1GB 2Rx8 PC2-4200M-544-11-ZZ
is a 1GB DDR2 Micro-DIMM using 2 ranks of x8 SDRAMs operational to PC2-4200 performance with CAS Latency = 5, t_{RCD} = 4, t_{RP} = 4, using JEDEC SPD revision 1.1, no JEDEC raw card reference design file standard used for the assembly

Examples, with font change:

2GB 2Rx16 PC2-5300S-444 11-A1
is a 2GB DDR2 SO-DIMM using 2 ranks of x16 SDRAMs operational to PC2-5300 performance with CAS Latency = 4, t_{RCD} = 4, t_{RP} = 4, using JEDEC SPD revision 1.1, raw card reference design file A revision 1 used for the assembly

512MB 1Rx8 PC2-6400U-555 11-A1
is a 512MB DDR2 Unbuffered DIMM with 64 bit wide data bus (no ECC) using 1 rank of x8 SDRAMs operational to PC2-6400 performance with CAS Latency = 5, t_{RCD} = 5, t_{RP} = 5, using JEDEC SPD revision 1.1, raw card reference design file A revision 1 used for the assembly.

512MB 1Rx8 PC2-6400E-555 11-A1
is a 512MB DDR2 Unbuffered DIMM with 72 bit wide data bus (ECC supported) using 1 rank of x8 SDRAMs operational to PC2-6400 performance with CAS Latency = 5, t_{RCD} = 5, t_{RP} = 5, using JEDEC SPD revision 1.1, raw card reference design file A revision 1 used for the assembly.

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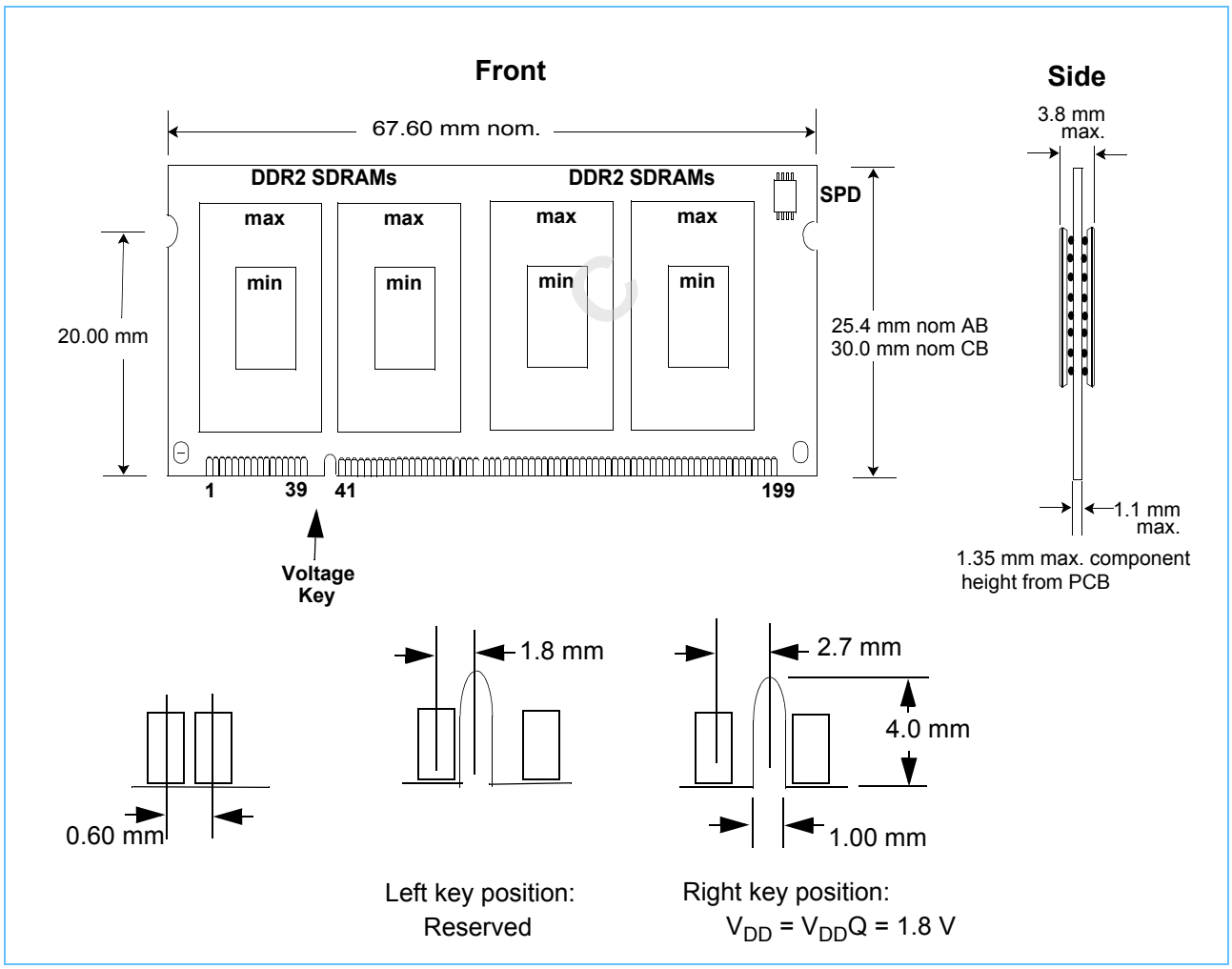
10. SO-DIMM Mechanical Specifications

JEDEC has standardized detailed mechanical information for the 200 Pin DIMM family. This information can be accessed on the worldwide web as follows:

1. Go to http://www.jedec.org.
2. Click on ‘Free Standards and Docs.’
3. Scroll down and double click on ‘Publication 95.’
4. Under ‘Outlines/Registrations,’ click on ‘Microelectronics Outlines.’
5. Scroll down and select ‘MO-224’ to download the PDF for this product family.

This example is for reference only; please refer to JEDEC standard MO-224 variation AB or CB for details.

Reference Simplified Mechanical Drawing with Keying Position



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